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DESIGN AND IMPLEMENTATION OF A SIDE-CHANNEL RESISTANT FPGA-BASED SIGNING ENGINE FOR CRYPTOCURRENCY HARDWARE WALLETS

A Research Paper

BY

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ABSTRACT

The cryptocurrency ecosystem recorded at least USD 154 billion in illicit transaction volume during 2025, alongside USD 3.4 billion in outright theft, exposing persistent weaknesses in the physical security of hardware wallet devices. Contemporary commercial hardware wallets rely on general-purpose Microcontroller Units (MCUs), which execute the SECP256K1 Elliptic Curve Digital Signature Algorithm (ECDSA) as a sequential software routine. This execution model produces observable power and electromagnetic emission profiles exploitable through Side-Channel Analysis (SCA). Separately, the February 2025 Bybit incident, in which more than USD 1.4 billion was drained without any cryptographic key being compromised, demonstrated that key confidentiality alone is an insufficient security property: the integrity of the signing authorisation path is equally critical.

This research addresses both failure modes through a single reconfigurable artefact. A constant-time SECP256K1 ECDSA signing engine is designed and implemented in VHDL on the AMD Artix-7 XC7A35T device hosted on a MicroPhase A7-Lite development board. The scalar multiplication datapath employs the Montgomery ladder over homogeneous projective coordinates using the complete addition formulas of Renes, Costello and Batina, which admit no exceptional cases and therefore require no data-dependent special-case logic. Coron projective re-randomisation and scalar blinding provide first-order Differential Power Analysis (DPA) resistance. A second contribution addresses the authorisation path: the engine renders the transaction destination and value directly to an HDMI output from within the same fabric that holds the private key, and requires physical button confirmation before a signature is released, structurally eliminating the host-mediated display substitution exploited at Bybit.

The research follows the Design-Science Research paradigm. Functional correctness of the field arithmetic and point arithmetic is established first against a Python reference model, validated against published RFC 6979 test vectors and cross-checked independently against a textbook affine implementation using modular inversion, a separate computation path from the one under test. The same instruction sequence the model executes for point addition and doubling is emitted directly as the hardware microcode, so model and hardware share one definition and cannot silently drift apart. The modular multiplier, modular adder/subtractor, point unit and Montgomery ladder are implemented in VHDL, verified first in simulation against this model (20 of 20 multiplier vectors and 24 of 24 point-arithmetic vectors passing, including the four degenerate cases a Jacobian implementation would need a branch to handle), and subsequently synthesised, placed and routed for the Artix-7 XC7A35T. The implemented signing core closes timing at 50 MHz with a worst negative slack of +0.405 ns, consuming 40.4 per cent of available Look-Up Tables and zero of the ninety DSP48E1 slices or fifty Block RAM tiles, and both point operations execute in a fixed, operand-independent number of clock cycles, measured directly on the synthesised design rather than predicted. The resulting

signature latency of 45.0 ms at 50 MHz exceeds the 25 ms design target; the shortfall is traced precisely to the bit-serial multiplier architecture, and a costed radix-4/radix-8 optimisation is presented, projected to close it using DSP resources confirmed free on the device. On-chip side-channel sensing, CPA and TVLA acquisition, the protocol controller and the trusted display remain outstanding and are reported as such.

Keywords: FPGA, ECDSA, SECP256K1, Side-Channel Analysis, Hardware Wallet, Montgomery Ladder, Complete Addition Formulas, TVLA, Artix-7, Constant-Time Cryptography, What-You-See-Is-What-You-Sign.

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LIST OF ABBREVIATIONS

ADC	Analogue-to-Digital Converter
ASIC	Application-Specific Integrated Circuit
BRAM	Block Random Access Memory
CPA	Correlation Power Analysis
DPA	Differential Power Analysis
DSP	Digital Signal Processing block
DSR	Design-Science Research
ECC	Elliptic Curve Cryptography
ECDSA	Elliptic Curve Digital Signature Algorithm
EDA	Electronic Design Automation
FPGA	Field Programmable Gate Array
FSM	Finite State Machine
HDL	Hardware Description Language
HDMI	High-Definition Multimedia Interface
HMAC	Hash-based Message Authentication Code
IOB	Input/Output Block
LUT	Look-Up Table
MCU	Microcontroller Unit
MMCM	Mixed-Mode Clock Manager
RCB	Renes-Costello-Batina complete addition formulas
RFC	Request for Comments
RNG	Random Number Generator
RO	Ring Oscillator
RTL	Register-Transfer Level
SCA	Side-Channel Analysis
SHA	Secure Hash Algorithm
SPA	Simple Power Analysis
TDC	Time-to-Digital Converter
TMDS	Transition-Minimised Differential Signalling
TRNG	True Random Number Generator
TVLA	Test Vector Leakage Assessment
UART	Universal Asynchronous Receiver-Transmitter
WNS	Worst Negative Slack
WYSIWYS	What-You-See-Is-What-You-Sign

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

The emergence of Bitcoin in 2009 catalysed the development of a multi-faceted digital asset ecosystem encompassing Decentralised Finance (DeFi), Non-Fungible Tokens (NFTs), and institutional-grade trading platforms [1]. Ownership of all assets recorded on these immutable, irreversible blockchain ledgers is established exclusively by the possession of cryptographic private keys [2]. Private key compromise constitutes a catastrophic and permanent financial loss, as there is no institutional recourse or transaction reversal mechanism available [3].

A comprehensive analysis of major wallet breaches from 2012 to 2025 documented 85 significant incidents resulting in losses exceeding USD 6.98 billion [4]. Chainalysis reports that illicit cryptocurrency addresses received at least USD 154 billion during 2025, representing a 162 per cent year-on-year increase driven principally by a 694 per cent rise in value received by sanctioned entities [5]. Of this, USD 3.4 billion was stolen outright. Actors linked to the Democratic People's Republic of Korea accounted for USD 2.02 billion of that total and were responsible for a record 76 per cent of all service compromises recorded during the year [5], [6]. Losses attributable to compromise of individual personal wallets totalled USD 713 million across approximately 80,000 victims [14].

Notwithstanding their superior security posture relative to software custody, commercial hardware wallets based on general-purpose microcontrollers retain fundamental physical security vulnerabilities at the cryptographic signing layer. The digital signature operation, specifically Elliptic Curve Scalar Multiplication (ECSM) within SECP256K1 ECDSA, is executed as a sequential software routine. The Complementary Metal-Oxide-Semiconductor (CMOS) switching activity associated with processing each secret key bit produces a power consumption profile that measurably reflects the Hamming weight of intermediate computation values [9], enabling Side-Channel Analysis attacks.

The exploitability of this vulnerability was demonstrated in 2019, when Ledger Donjon researchers extracted the recovery seed from a Trezor One hardware wallet in under five minutes using approximately USD 100 of equipment, and characterised the flaw as unfixable without hardware redesign [10]. In 2023, security firm Unciphered intercepted the unencrypted inter-chip communication channel between the microprocessor and Secure Element of the OneKey wallet using an FPGA-based logic analyser, extracting the mnemonic seed phrase in transit [11]. Ledger Donjon subsequently reported in 2024 that cryptographic operations on the Trezor Safe 3 could still be induced to execute on the general-purpose microcontroller rather than exclusively within the certified secure element, re-opening the same class of exposure on a current-generation device [36].

A second and architecturally distinct failure mode became prominent in 2025. On 21 February of that year, more than USD 1.4 billion in assets, including 401,347 ETH, was drained from a Bybit cold wallet in the largest cryptocurrency theft ever recorded [37]. Forensic analysis established that no private key was compromised and that no defect existed in the underlying smart contracts. The attackers instead injected malicious JavaScript into the wallet management front-end accessed by the authorised signers, causing the interface to display a benign transaction while the signers in fact authorised a modification to the wallet's controlling logic [37], [38]. The cryptography performed exactly as specified; the human authorisation decision was subverted because the display could not be trusted.

This research responds to both failure modes. It proposes an FPGA-based signing engine implemented on the AMD Artix-7 XC7A35T device, which enforces constant-time SECP256K1 ECDSA signing, eliminates the inter-chip communication attack class by confining all sensitive operations within a single fabric boundary, and renders the transaction under authorisation directly to a display driven by that same fabric. The EthVault project established the feasibility of consolidating a complete wallet cryptographic stack into reconfigurable logic, achieving a full SECP256K1 and BIP-32 architecture within 27 per cent of the available Look-Up Tables of a Zynq UltraScale+ device [12].

Figure 1.1 - Problem Identification Flowchart: Dual Attack Surfaces in MCU-Based Hardware Wallets.

1.2 Statement of the Problem

The central problem addressed by this research is twofold, and prior work has consistently treated only the first half of it.

The first component is the persistence of physical side-channel vulnerabilities in the ECDSA signing core of MCU-based hardware wallets. Software implementations of ECDSA on microcontrollers produce non-uniform execution profiles at the point multiplication step. Data-dependent control flow, cache behaviour and branch prediction produce a power trace whose statistical features correlate with the Hamming weight of the operands being processed, enabling Differential Power Analysis to reconstruct private keys from captured traces [9], [10]. Research confirms that private key compromise accounts for approximately 70 per cent of cryptocurrency theft events [13]. A closely related exposure exists in segmented designs where a microprocessor communicates with a dedicated Secure Element over an unencrypted printed circuit board data bus, as demonstrated by the 2023 OneKey exploit [11].

The second component is the integrity of the authorisation path. A hardware wallet that never discloses its key is nonetheless insecure if the human operator cannot verify what they are authorising. The Bybit incident establishes that this is not a theoretical concern: an attacker who controls the display can obtain a valid signature over an arbitrary transaction while the signer believes they are approving something else [37], [38]. Commercial devices mitigate this

partially through small on-device screens, but the rendering path in such devices typically traverses firmware that is itself a target, and the certification claims attaching to secure elements do not extend to the display subsystem or to the companion application.

These two problems share a common structural cause. In the conventional architecture, the signing computation and the authorisation display are both mediated by general-purpose software running on a processor that an attacker may influence. A monolithic reconfigurable implementation, in which the signing datapath is fixed logic and the display is driven from the same fabric, addresses both simultaneously. The financial motivation is severe: individual victim losses in 2025 totalled USD 713 million [14], while state-linked actors illicitly obtained USD 2.02 billion through targeted exploitation of key management and signing infrastructure [5].

1.3 Aim and Objectives

Aim: To design, implement and evaluate a side-channel resistant SECP256K1 ECDSA signing engine with a trusted transaction display on a low-cost AMD Artix-7 FPGA platform, demonstrating improved protection against both power-based side-channel analysis and authorisation-path substitution relative to commercial MCU-based hardware wallets.

Objectives:

- **Objective 1:** Characterise side-channel and authorisation-path vulnerabilities in commercial hardware wallets through systematic review of documented attacks, including the 2019 Trezor One power analysis, the 2023 OneKey bus interception and the 2025 Bybit authorisation compromise [10], [11], [37].
- **Objective 2:** Design a constant-time SECP256K1 signing engine in VHDL employing the Montgomery ladder over homogeneous projective coordinates with the complete addition formulas of Renes, Costello and Batina, incorporating register-transfer-level countermeasures against differential power analysis.
- **Objective 3:** Synthesise the signing core for the Artix-7 XC7A35T device, targeting resource utilisation within the 20,800 available Look-Up Tables, and characterise the achievable area against latency trade-off for the modular multiplier by comparing the implemented bit-serial architecture against higher-radix alternatives.
- **Objective 4:** Evaluate signing latency against real-world blockchain requirements, targeting a signature latency below 25 ms.
- **Objective 5:** Validate side-channel resistance using an on-chip voltage-drop sensor, cycle-accurate constant-time verification and simulation-based leakage assessment, targeting a Test Vector Leakage Assessment statistic below the conventional threshold of 4.5 while demonstrating that the same measurement chain detects leakage in a deliberately unprotected control implementation.

- **Objective 6:** Implement a trusted transaction display in which the destination address and transfer value are rendered to an HDMI output directly from the signing fabric and a signature is released only upon physical button confirmation, structurally mitigating the authorisation substitution class exemplified by the Bybit incident.

1.4 Research Questions and Hypothesis

Research Questions:

- RQ1: How does an FPGA-based constant-time signing architecture compare to MCU-based wallet implementations in resistance to differential and simple power analysis, as quantified by Test Vector Leakage Assessment statistics obtained from an on-chip voltage-drop sensor?
- RQ2: What register-transfer-level architectural choices are required to implement a complete SECP256K1 ECDSA signing engine within the 20,800 Look-Up Table budget of an Artix-7 XC7A35T device, and what area against latency trade-off do they impose?
- RQ3: To what extent does a monolithic FPGA signing engine mitigate the inter-chip communication vulnerabilities present in dual-chip microcontroller and secure element architectures [11]?
- RQ4: To what extent does rendering the transaction under authorisation from within the signing fabric mitigate the authorisation substitution attack class demonstrated at Bybit [37]?
- RQ5: Can a defensible side-channel evaluation be conducted using only on-chip instrumentation, without an oscilloscope or dedicated acquisition platform?

Hypothesis:

H1: The proposed FPGA signing engine will exhibit a Test Vector Leakage Assessment Welch t-statistic below 4.5 across the acquired trace population during SECP256K1 scalar multiplication, under conditions in which the same acquisition chain yields a statistic exceeding 4.5 for an unprotected control implementation, while achieving a signature latency below 25 ms on the Artix-7 XC7A35T.

1.5 Significance and Scope

This research makes contributions to four stakeholder groups. For cryptocurrency users, it provides hardware-level signing isolation combined with a trusted authorisation display, addressing the two attack classes responsible for the majority of recorded losses. For the hardware wallet manufacturing industry, it offers a validated signing engine design with published benchmarks obtained on a commodity device costing under USD 150. For the academic community, it establishes reproducible performance and security benchmarks for constant-time SECP256K1 ECDSA on resource-constrained 7-series devices, an area in which prior work has targeted substantially larger and more expensive platforms. For regulators, the

European Union Markets in Crypto-Assets Regulation and the Digital Operational Resilience Act mandate technical security baselines for digital asset custody, applicable to service providers from December 2024 and January 2025 respectively.

A distinct methodological contribution arises from the resource constraints under which this work was conducted. The evaluation methodology developed in Chapter 3 permits side-channel assessment of an FPGA cryptographic implementation using only the target device itself, free tooling and open-source instrumentation. This lowers the barrier to hardware security research in institutions where dedicated acquisition platforms are not available, a condition that characterises the majority of engineering departments in developing economies.

The scope is bound to SECP256K1 ECDSA signing with externally supplied message digests and BIP-32 derived key material, implementation on the Artix-7 XC7A35T, and side-channel evaluation using on-chip sensing, timing analysis and simulation-based assessment. Post-quantum signature schemes, multi-signature protocols, BIP-39 mnemonic generation and management, and companion mobile applications are designated as future work.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter establishes the theoretical and empirical foundation upon which the proposed FPGA signing engine is justified, and identifies the specific research gaps this project addresses. With state-linked actors responsible for 76 per cent of service compromises during 2025 [5], and a systematic review of the preceding decade documenting over USD 5.4 billion in wallet-related losses [4], the urgency of the problem is empirically established. The review progresses from foundational cryptographic concepts and side-channel theory, through commercial wallet analysis and the emerging authorisation-path threat class, to FPGA prototypes, evaluation methodology under instrumentation constraints, a five-year empirical review, and finally the gap analysis.

2.2 Theoretical Framework

2.2.1 Cryptographic Foundations

The security of Bitcoin and Ethereum is grounded in Elliptic Curve Cryptography over a prime field [21]. The SECP256K1 curve, defined by $y^2 = x^3 + 7$ over the prime field F_p with $p = 2^{256} - 2^{32} - 977$, underpins both networks. A private key k is a randomly generated integer in the interval [1, n), from which the public key Q is derived as $Q = kG$, where G is the publicly known generator point and n is the group order. Security depends on the intractability of the Elliptic Curve Discrete Logarithm Problem; the most efficient known generic algorithm requires $O(\sqrt{n})$ steps, providing approximately 128 bits of classical security for 256-bit parameters [21]. The BIP-32 hierarchical deterministic framework enables derivation of unlimited child key pairs from a single master seed using HMAC-SHA-512 [22].

The nonce k used in each ECDSA signature warrants particular attention. If even a small number of bits of the nonce leak across a sufficient number of signatures, lattice reduction techniques recover the private key outright. This makes nonce handling the single most security-critical element of a signing implementation, and motivates the adoption of deterministic nonce derivation per RFC 6979 [39] in preference to reliance on a hardware random number generator alone.

2.2.2 Side-Channel Attack Theory

Side-Channel Analysis exploits physical emissions that correlate with secret data, rather than attacking the mathematical structure of the algorithm [9]. In CMOS circuits, dynamic power consumption is proportional to the number of bit transitions occurring in a given clock cycle, so a power trace carries information about the values being manipulated. Two attack classes are relevant. Simple Power Analysis examines a single trace to distinguish point additions from doublings within scalar multiplication and thereby infer key bits directly [10]. Differential

Power Analysis applies statistical analysis across many traces, typically using the Pearson correlation coefficient against a Hamming weight or Hamming distance power model, to identify key-dependent patterns even under significant measurement noise [9]. For hardware implementations, the Hamming distance model, which reflects transitions between consecutive register states, is generally the appropriate choice.

Scalar multiplication presents a different attack surface from block ciphers. There is no small key byte over which an adversary can enumerate hypotheses; attacks instead target individual scalar bits within each ladder iteration, or profile intermediate point coordinates. Consequently horizontal attacks, which exploit the many samples within a single trace, and single-trace template attacks remain effective against implementations protected only by countermeasures that randomise across traces.

The Test Vector Leakage Assessment methodology introduced by Goodwill et al. [18] applies Welch's t-test to compare traces acquired under a fixed input against traces acquired under random inputs. A statistic exceeding 4.5 in magnitude at any sample point indicates statistically significant first-order leakage. Subsequent work has qualified this threshold: with very long traces the multiple comparisons problem causes a leakage-free device to be declared leaky with high probability at a fixed threshold of 4.5, and the threshold should properly scale with trace length [40]. Standaert has further cautioned that a pass or fail verdict under this test may be only loosely connected to the actual security level of a masked design [41]. These qualifications inform the evaluation protocol adopted in Section 3.7.

2.3 Wallet Classifications and Commercial Landscape

Cryptocurrency wallets are classified along three principal dimensions: connectivity, custody model, and form factor [7].

Dimension	Category A	Category B
Connectivity	Hot wallet (online, e.g. MetaMask): high convenience, large attack surface	Cold wallet (offline, air-gapped): reduced attack surface
Custody	Custodial (exchange-managed): risk concentrated at the provider [13]	Non-custodial (self-managed): risk transfers to the user
Form factor	Software (application or browser extension): exposed to host operating system vulnerabilities	Hardware (Trezor, Ledger): keys do not leave the device under normal operation

Table 2.1 - Cryptocurrency Wallet Classification Matrix.

Figure 2.1 - Cryptocurrency Wallet Classification Taxonomy, ordered by security level.

The hardware wallet market is dominated by two architectural paradigms: general-purpose microcontroller designs, as employed by Trezor using STM32-series devices, and hybrid designs pairing a microcontroller with a certified Secure Element, as employed by Ledger [7],

[24]. Trezor open-source firmware enables community auditing but has been conclusively demonstrated vulnerable to power analysis [10]. Ledger Secure Elements provide certified physical tamper resistance, but the ECDSA implementation executing within the secure element remains a software routine on a microprocessor core, preserving the sequential execution model that gives rise to data-dependent emissions.

A frequent misunderstanding warrants explicit treatment. Common Criteria certifications advertised at EAL5+ or EAL6+ apply to the secure element integrated circuit evaluated against a defined protection profile. They do not certify the wallet firmware, the transaction display path, the companion application, or the product as a whole. The Bybit incident is the definitive illustration: certified components were present throughout and the loss occurred regardless, because the compromised element lay outside the certified boundary [37].

Market sizing for this sector is reported inconsistently across research firms, and figures should be treated with caution. IMARC Group estimates the hardware wallet market at USD 564.6 million in 2025, projected to reach USD 2,648.3 million by 2034 at a compound annual growth rate of 18.17 per cent. Mordor Intelligence estimates USD 0.54 billion in 2025 rising to USD 2.25 billion by 2031 at 25.6 per cent. The two agree closely on the present size while differing materially on the growth trajectory. What is not disputed is the direction: sustained double-digit growth driven by institutional adoption and by the regulatory requirements discussed in Section 1.5.

Figure 2.2 - Architecture Comparison: MCU-Based Wallet (left) versus Proposed FPGA Signing Engine (right), illustrating elimination of the inter-chip sniffing attack surface and of the untrusted display path.

2.4 Vulnerability Analysis

A systematic literature review [4] categorises physical attack vectors across six principal surfaces. Table 2.2 summarises the most critical documented vulnerabilities together with the corresponding mitigations offered by the proposed architecture.

Attack type	Target	Documented case	FPGA mitigation
Differential and simple power analysis	Scalar multiplication on MCU	Trezor One, 2019 [10]	Constant-time datapath; Montgomery ladder with complete formulas; Coron re-randomisation
Voltage glitching	Read-protection bypass	Trezor One, 2020; Trezor Safe 3, 2024 [36]	Hardened I/O constraints; bitstream authentication
Inter-chip bus interception	MCU to secure element bus	OneKey, 2023 [11]	Monolithic fabric; no external key-bearing bus
Firmware injection	Bootloader vulnerabilities	Ledger Nano S [19]	Bitstream authentication; fixed logic
Supply chain substitution	Pre-seeded keys	USB Armory counterfeit wallet [19]	On-chip TRNG; auditable bitstream

Attack type	Target	Documented case	FPGA mitigation
Authorisation substitution	Signer display path	Bybit, 2025 [37], [38]	Transaction rendered from signing fabric; physical confirmation required

Table 2.2 - Critical Vulnerability Summary and Proposed FPGA Signing Engine Mitigations.

The 2023 OneKey exploit [11] provides a particularly instructive case study for the first group of attacks. By deploying an FPGA as a high-speed logic analyser on the printed circuit board traces between the wallet microprocessor and its secure element, researchers captured the unencrypted BIP-39 mnemonic seed phrase in transit. This demonstrates that a formally certified secure element provides no protection against attacks on the unencrypted communication channel feeding it. A monolithic signing engine, in which all cryptographic operations share the same reconfigurable fabric and communicate exclusively through internal logic, structurally eliminates this vector [12].

Figure 2.3 - Chronological Timeline of Hardware Wallet Security Incidents (2018-2025).

2.5 The Authorisation Path as a Distinct Attack Surface

The literature on hardware wallet security has focused overwhelmingly on key confidentiality. The systematic reviews of Houy et al. [4] and Erinle et al. [13] both organise their taxonomies around mechanisms by which key material may be extracted or substituted. This emphasis is historically justified, but it is incomplete, and the events of 2025 have made the omission consequential.

On 21 February 2025 more than USD 1.4 billion was drained from a Bybit cold wallet, comprising 401,347 ETH valued at approximately USD 1.12 billion together with staked and wrapped Ether derivatives [37]. Independent forensic analyses converge on the mechanism. Malicious JavaScript was introduced into the wallet management interface served to the authorised signers, configured to activate only under specific conditions. When the signers reviewed the pending transaction, the interface presented a routine transfer. The transaction actually placed before them for signature modified the controlling logic of the multi-signature wallet, transferring effective control to the attacker [37], [38]. Every signature produced was cryptographically valid. Every key remained confidential. The audit of the underlying contracts found no defect.

The security property violated is what the hardware security community terms What-You-See-Is-What-You-Sign. Formally, it requires that the representation of a transaction presented to the human authoriser be derived from the same data that will be signed, through a path the adversary cannot influence. Commercial hardware wallets address this partially by rendering transaction details on a small integrated display, on the reasoning that a compromised host cannot alter what the device itself draws. This reasoning is sound in principle. In practice the rendering path in such devices passes through device firmware, which is updatable, complex,

and has itself been the subject of documented vulnerabilities [19], [36]; and where a transaction is too complex for the device to parse and display meaningfully, users are presented with an opaque hash and are effectively signing blind.

An FPGA implementation offers a structurally stronger position. If the display controller is fixed logic within the same fabric as the signing datapath, and the values rendered are taken from the identical registers that feed the hash under signature, then the rendering path contains no mutable software and no external bus. The property is enforced by the topology of the circuit rather than by the correctness of a program. To the best of the author's knowledge, no published FPGA wallet prototype has implemented or evaluated this property, a gap identified explicitly in Section 2.9.

Figure 2.4 - Threat Model: Key Extraction versus Authorisation Substitution, showing the disjoint mitigation requirements of the two attack classes.

2.6 FPGA-Based Research Prototypes

The EthVault project [12] represents the most directly relevant prior work. Implementing a complete SECP256K1 architecture with BIP-32 and BIP-44 hierarchical deterministic key derivation on a Xilinx Zynq UltraScale+ device, the design occupies 62,209 Look-Up Tables, corresponding to 27 per cent of the target device, with 31,180 registers and no DSP blocks. A single scalar multiplication requires 1,887,520 clock cycles, yielding approximately 11.3 ms at 167 MHz. The architecture employs the Montgomery ladder with a temporary register arrangement ensuring both accumulator registers are accessed regardless of the scalar bit, and adopts the complete addition formulas of Renes, Costello and Batina [42] rather than conventional Jacobian arithmetic.

EthVault is significant for the present work in a further respect. The authors additionally synthesised the SECP256K1 module alone for an Artix-7 device, reporting approximately 24,000 Look-Up Tables at 90 MHz. This figure exceeds the entire 20,800 Look-Up Table capacity of the XC7A35T targeted here, and establishes directly that a port of that architecture is infeasible on the present platform. A more area-efficient datapath is therefore required, which motivates the bit-serial multiplier design developed in Section 3.13.

The wider literature confirms that such a design is achievable. Table 2.3 summarises reported utilisation for prime-field scalar multiplication implementations on 7-series and comparable devices.

Work	Device	Resources	Frequency	Scalar multiplication latency
Hao et al. [43]	Kintex-7	6,477 slices	156.3 MHz	1.73 ms
Hao et al. [43]	Zynq-7000	6,351 slices	149.7 MHz	1.80 ms

Work	Device	Resources	Frequency	Scalar multiplication latency
Islam et al. [44]	Virtex-7	8,873 slices	177.7 MHz	1.48 ms
Nguyen et al. [45]	Artix-7	6,471 slices; 23,187 LUT; 8 BRAM	110.4 MHz	1.48 ms (5.92 ms full ECDSA)
EthVault [12]	Artix-7 (SECP256K1 module)	~24,000 LUT	90 MHz	20.97 ms
This work	Artix-7 XC7A35T (point unit, demo_top)	8,401 LUT (40.4%); 0 DSP; 0 BRAM	50 MHz	45.0 ms (derived, full protected signature)

Table 2.3 - Reported Resource Utilisation of Prior Prime-Field ECC Implementations.

The designs of Hao et al. [43] at approximately 6,400 slices are the strongest evidence that a complete scalar multiplication engine fits within the 8,150 slices available on the XC7A35T, and the present work corroborates this directly: the synthesised point unit measures 8,401 LUTs, consistent with that literature figure and confirming the area feasibility argued in Section 2.6 with a hardware measurement rather than a projection. Complementary work by Poncha et al. [25] and Ay [26] provides additional benchmarks for 256-bit field multiplication and constant-time accelerator design directly applicable to the arithmetic unit developed in this project. Conversely, high-throughput designs reported in the literature consume 52,800 Look-Up Tables and 54 DSP blocks, or in other cases between 560 and 1,036 DSP blocks, which places them far outside the 90 DSP48E1 budget of the present device and confirms that throughput-optimised architectures are not portable to this class of platform. The present design occupies the opposite end of that trade-off from the high-throughput group: it is markedly slower than the dedicated ECC processors in the table, a direct consequence of the bit-serial multiplier documented in Section 3.13, and Section 4.7 characterises the cost of narrowing that gap.

2.7 Side-Channel Evaluation Without Dedicated Instrumentation

Conventional side-channel evaluation requires an oscilloscope or a dedicated acquisition platform such as ChipWhisperer, together with a shunt resistor inserted into the power rail of the target device. Neither the acquisition hardware nor a board providing a power measurement point was available for this project, a constraint discussed candidly in Section 3.2. A body of literature exists, however, that removes this dependency entirely.

Schellenberg et al. [46] demonstrated that a delay sensor instantiated within the FPGA fabric can observe the voltage fluctuations induced by a co-located cryptographic circuit through the shared power distribution network, and can recover key material without any external measurement apparatus. Zhao and Suh [47] independently established the same result using ring-oscillator based power monitors, recovering an RSA key on a Zynq-7000 device whose

programmable logic is the same fabric family as the Artix-7 used here. Gravellier et al. [48] refined the ring-oscillator sensor for higher acquisition rates.

The sensor architecture most commonly adopted is a Time-to-Digital Converter constructed from a carry-chain delay line sampled by a bank of flip-flops. The propagation depth reached by a clock edge within a fixed interval depends on supply voltage, so the thermometer-coded output tracks voltage droop and therefore the switching activity of neighbouring logic. Spielmann, Glamočanin and Stojilović [49] subsequently introduced a routing delay sensor which achieves key recovery with, on average, 35 per cent fewer traces than a carry-chain converter while requiring no strict placement or routing constraints, substantially reducing the implementation burden. Open-source register-transfer-level implementations of both sensor types targeting AMD 7-series devices are publicly available, together with the placement constraints required.

Two complementary methods further strengthen an instrument-free evaluation. First, constant-time behaviour can be verified directly by instrumenting the design with a cycle counter and applying the statistical methodology of Reparaz et al. [50] to the resulting timing distributions. For a correctly designed hardware datapath the cycle count should be exactly invariant with respect to the secret scalar, which is a stronger and more readily falsifiable claim than a statistical bound. Second, simulation-based leakage assessment derives estimated power traces from switching activity captured during post-implementation simulation, permitting Test Vector Leakage Assessment to be applied before any measurement is taken [51]. Neither method substitutes for physical measurement, but in combination with an on-chip sensor they constitute a defensible evaluation.

One methodological requirement is emphasised because it is frequently omitted. An evaluation reporting an absence of leakage is uninterpretable unless the same acquisition chain is shown capable of detecting leakage that is known to be present. This project therefore retains a deliberately unprotected implementation of the signing engine within the same bitstream as a positive control, following standard practice in leakage detection [18].

2.8 Empirical Review of Prior Quantitative Studies

2.8.1 Selection Rationale

The following review synthesises six studies published between 2020 and 2025, selected to trace the evolution of academic and industrial understanding of wallet security. Each is assessed against three criteria: the approach adopted, the measurable achievement reported, and the residual gap motivating subsequent work.

2.8.2 2020 - TEE-Based Key Isolation

Cerdeira, Santos, Fonseca and Pinto [35] conducted a security analysis of five commercial TrustZone-assisted Trusted Execution Environment implementations, reverse-engineering firmware and analysing over one hundred publicly reported vulnerabilities. All five had suffered exploitable defects, and four of five exposed a side-channel leakage vector through cache resources shared between the normal and secure worlds. The residual gap is that the study did not evaluate physical power side-channel leakage during signing. A trusted execution environment is an isolation mechanism, not a constant-time execution guarantee; the secure world executes on the same physical core and produces the same data-dependent emissions. This gap motivates hardware-level implementation directly.

2.8.3 2021 - Supply-Chain and Architectural Attacks

Dabrowski et al. [19] performed a systematic architectural analysis of Trezor, Ledger and KeepKey devices focusing on initialisation and authenticity verification, and constructed a functional counterfeit wallet from a USB Armory single-board computer that passed all authenticity checks of the official client software while generating key material under attacker control. Three supply-chain vectors were demonstrated: key replacement via a pre-initialised device, firmware replacement through bootloader bypass, and clone attacks using USB traffic replay. The residual gap is that the proposed mitigation, a modified collaborative key generation architecture, addresses provisioning but leaves the physical signing operation unchanged.

2.8.4 2022 - Physical Side-Channel Escalation

Work in this period consolidated the practical accessibility of physical attacks, reducing the equipment cost and expertise required to extract key material from consumer devices. The residual gap is the absence of a defended reference implementation against which countermeasure efficacy could be measured on comparable hardware.

2.8.5 2023 - Systematic Literature Review

Houy, Schmid and Bartel [4] surveyed the security aspects of cryptocurrency wallets across the preceding decade, cataloguing 85 significant incidents and losses exceeding USD 6.98 billion, and establishing a vulnerability taxonomy that has become a standard reference. The residual gap, evident with the benefit of hindsight, is that the taxonomy is organised around key compromise and does not accommodate attacks in which correct keys are used to authorise incorrect transactions.

2.8.6 2025 - FPGA Wallet Prototype

Lemayian, Gagnon, Zhang and Giard [12] implemented a complete Ethereum cold wallet on a Zynq UltraScale+ device, achieving 27 per cent Look-Up Table utilisation and approximately 11.3 ms signature latency, with a reported mean squared error of 0.001840 between power

traces captured under two distinct keys as evidence of uniform execution. The residual gap is that no formal leakage assessment was reported. Neither Correlation Power Analysis nor Test Vector Leakage Assessment results were published, and no unprotected control was evaluated, so the quantitative side-channel resistance of the design remains open. This is the gap that Objective 5 of the present work addresses.

2.8.7 2025 - The Bybit Incident

The Bybit compromise [37], [38] is included here not as a peer-reviewed study but as the largest recorded empirical data point in the field. Its significance for the literature is that it falsifies the implicit assumption underlying the preceding six years of work, namely that securing the key secures the asset. The residual gap it exposes is the absence of any published hardware architecture that enforces What-You-See-Is-What-You-Sign at the level of circuit topology. Objective 6 addresses this gap.

Table 2.4 consolidates the review.

Year	Study	Achievement	Residual gap
2020	Cerdeira et al. [35]	Vulnerability taxonomy for TrustZone TEEs; cache side channel in four of five implementations	No physical power analysis of signing; isolation is not constant-time execution
2021	Dabrowski et al. [19]	Three supply-chain vectors demonstrated; functional counterfeit wallet constructed	Signing operation itself unchanged
2023	Houy et al. [4]	85 incidents catalogued; standard vulnerability taxonomy established	Taxonomy assumes key compromise as the loss mechanism
2025	Lemayian et al. [12]	Complete FPGA wallet at 27% LUT and 11.3 ms latency	No CPA or TVLA reported; no unprotected control
2025	Bybit incident [37]	Largest recorded loss, USD 1.4 billion, with all keys intact	No hardware architecture enforcing trusted display

Table 2.4 - Empirical Review: Approach, Achievement, and Gap (2020-2025).

2.9 Research Gaps

Four gaps emerge from the foregoing review and define the contribution of this research.

- **Gap 1:** No published FPGA hardware wallet implementation reports quantitative side-channel leakage assessment. EthVault [12], the closest prior work, reports a trace similarity metric but neither Correlation Power Analysis nor Test Vector Leakage Assessment results, and evaluates no unprotected control.
- **Gap 2:** Prior FPGA implementations target device classes an order of magnitude larger and more costly than commodity hardware. The area-efficiency question, namely whether a complete constant-time signing engine fits within a device of approximately 20,000 Look-Up Tables, is unresolved and is directly relevant to manufacturability.

- **Gap 3:** No published wallet architecture enforces What-You-See-Is-What-You-Sign at the level of circuit topology, notwithstanding that the largest loss on record arose from violation of precisely this property.
- **Gap 4:** Side-channel evaluation methodology in the wallet literature presumes access to dedicated acquisition instrumentation, which excludes the majority of institutions in developing economies from contributing to this field.

2.10 Summary

This chapter has established that hardware wallet security is presently constrained by two distinct and independently sufficient failure modes. The first is physical side-channel leakage arising from sequential software execution of scalar multiplication, demonstrated repeatedly against commercial devices and unaddressed by secure element certification. The second is substitution of the authorisation display, demonstrated at unprecedented scale in 2025 and structurally outside the scope of every countermeasure directed at key confidentiality. The review has further established that a constant-time signing engine of the required area is achievable on the target device class based on published implementations, that the architecture of the closest prior work is not portable to it, and that a credible side-channel evaluation may be conducted using on-chip instrumentation alone. Chapter 3 develops the methodology by which these findings are operationalised.

CHAPTER 3: DESIGN METHODOLOGY

3.1 Introduction and DSR Framework

This chapter details the methodology employed to design, implement and evaluate the proposed FPGA-based SECP256K1 ECDSA signing engine. The Design-Science Research paradigm, established by Hevner et al. [15] and operationalised through the six-phase process model of Peffers et al. [16], is adopted as the primary methodological framework. Three grounds justify this choice over alternative paradigms. First, Design-Science Research generates prescriptive knowledge through the construction of novel artefacts [15]. Second, hardware security vulnerabilities are frequently latent in physical implementations that pass all logical simulations, and the iterative evaluation loops of the paradigm accommodate the refinement cycle in which side-channel results feed back into register-transfer-level redesign [16]. Third, the documented exploits of commercial wallets [10], [11], [37] and the quantified scale of losses [5], [6] constitute the real-world problem context that drives the design objectives.

Figure 3.1 - Design-Science Research Methodology Framework, adapted from Peffers et al. [16] and Hevner et al. [15].

3.2 Implementation Platform and Deviation from the Specified Platform

The research originally specified implementation on the Avnet Ultra96-V2 development board, hosting a Xilinx Zynq UltraScale+ MPSoC (XCZU3EG-SBVA484), with side-channel evaluation conducted using a ChipWhisperer acquisition platform. That specification was selected because the device class corresponds directly to the EthVault benchmark platform [12] and because the integrated ARM application processor permits software-side test orchestration on the same silicon.

Neither the specified board nor the acquisition platform could be procured within the time and budget available for this research. The implementation reported here was therefore carried out on a MicroPhase A7-Lite development board hosting an AMD Artix-7 XC7A35T-2FGG484L device, and the side-channel evaluation methodology was reconstructed around on-chip instrumentation as described in Section 3.7. This deviation is reported here rather than in a concluding limitations section because it materially shapes every subsequent design decision and the reader requires it in advance. Table 3.1 states the difference precisely.

Parameter	Specified (Ultra96-V2)	Realised (A7-Lite)
Device	Zynq UltraScale+ XCZU3EG	Artix-7 XC7A35T-2FGG484L
Look-Up Tables	70,560	20,800
Flip-flops	141,120	41,600
DSP slices	216 (DSP48E2)	90 (DSP48E1)
Block RAM	216 tiles (36 Kb)	50 tiles (36 Kb); 1,800 Kb total
Hard processor	Quad-core ARM Cortex-A53	None; soft core or fixed-logic controller

Parameter	Specified (Ultra96-V2)	Realised (A7-Lite)
System clock	Programmable	50 MHz fixed oscillator
Host interface	ARM PS with Ethernet and USB	CH340 USB-UART bridge
Display output	DisplayPort via PS	HDMI transmit, 1080p at 60 Hz
SCA acquisition	ChipWhisperer with shunt	On-chip TDC sensor; no shunt available
Approximate cost	USD 250 plus USD 350 acquisition	Under USD 150 total

Table 3.1 - Specified versus Realised Implementation Platform.

Three consequences follow, each of which is addressed in the design rather than merely acknowledged.

The available Look-Up Table budget falls to less than one third of that specified. As established in Section 2.6, the EthVault SECP256K1 module alone requires approximately 24,000 Look-Up Tables on Artix-7, exceeding the entire capacity of the present device. A directly comparable architecture is therefore not merely inadvisable but impossible, and a bit-serial arithmetic datapath trading latency for area becomes mandatory. Section 3.13 develops this datapath and Section 4.7 characterises the resulting latency cost, together with a costed path to recover it.

The absence of a hard processor removes the option of software-side orchestration on the same die. Test sequencing, key loading and result retrieval are instead handled by a fixed-logic controller communicating with a host over the UART interface, with the host executing the reference model described in Section 3.12. This arrangement has an incidental methodological advantage: because the controller is fixed logic rather than software, the timing of the signing operation is not perturbed by operating system scheduling, which is a known source of jitter in processor-orchestrated acquisition.

The unavailability of acquisition instrumentation required the side-channel methodology to be reconstructed. Section 3.7 sets out the replacement in full.

The trade-off is worth stating plainly. What is lost is direct comparability with EthVault on identical silicon, and the higher signal quality obtainable from an external oscilloscope with a properly instrumented shunt. What is gained is a demonstration that the design fits a device costing under one tenth as much, and an evaluation methodology reproducible by any institution possessing the board alone. Given that manufacturability at low unit cost is a stated objective of the wallet industry, and that the exclusion of resource-constrained institutions from hardware security research is a recognised structural problem in the discipline, the revised platform is defensible on its own terms and not solely as a concession. The research questions in Section 1.4 were reformulated accordingly, and RQ5 exists specifically because of this constraint.

3.3 DSR Phase Mapping

DSR phase	Activity in this research	Output
1. Problem identification	Analysis of side-channel exploits on Trezor [10] and OneKey [11]; analysis of authorisation substitution at Bybit [37]; review of theft statistics [5], [6]	Problem statement (Section 1.2)
2. Objective specification	Definition of constant-time targets, leakage assessment pass criteria, and area budget for the XC7A35T	Objectives and H1 (Section 1.3)
3. Design and development	Reference model construction; VHDL implementation; synthesis; countermeasure integration	Reference model and bitstream (Chapter 3)
4. Demonstration	Deployment on the A7-Lite; functional signing verified against RFC 6979 and NIST vectors; live testnet transaction	Working prototype (Chapter 3)
5. Evaluation	On-chip sensor acquisition; TVLA and CPA against protected and unprotected variants; timing analysis; utilisation and timing reports	Security and performance data (Chapters 5 and 6)
6. Communication	Report documentation; publication of acquisition parameters and reference model for reproducibility	This report

Table 3.2 - DSR Phase Mapping for the FPGA Signing Engine Research.

3.4 Signing Engine Architecture Design

The signing engine employs a three-tier hierarchical architecture separating cryptographic operations across finite field arithmetic, elliptic curve point operations, and ECDSA protocol management. All sensitive data, including the private key scalar, intermediate field arithmetic operands and the signing nonce, remains confined within the fabric boundary at all times. No key-bearing signal is routed to a package pin, a property enforced during implementation by explicit constraint.

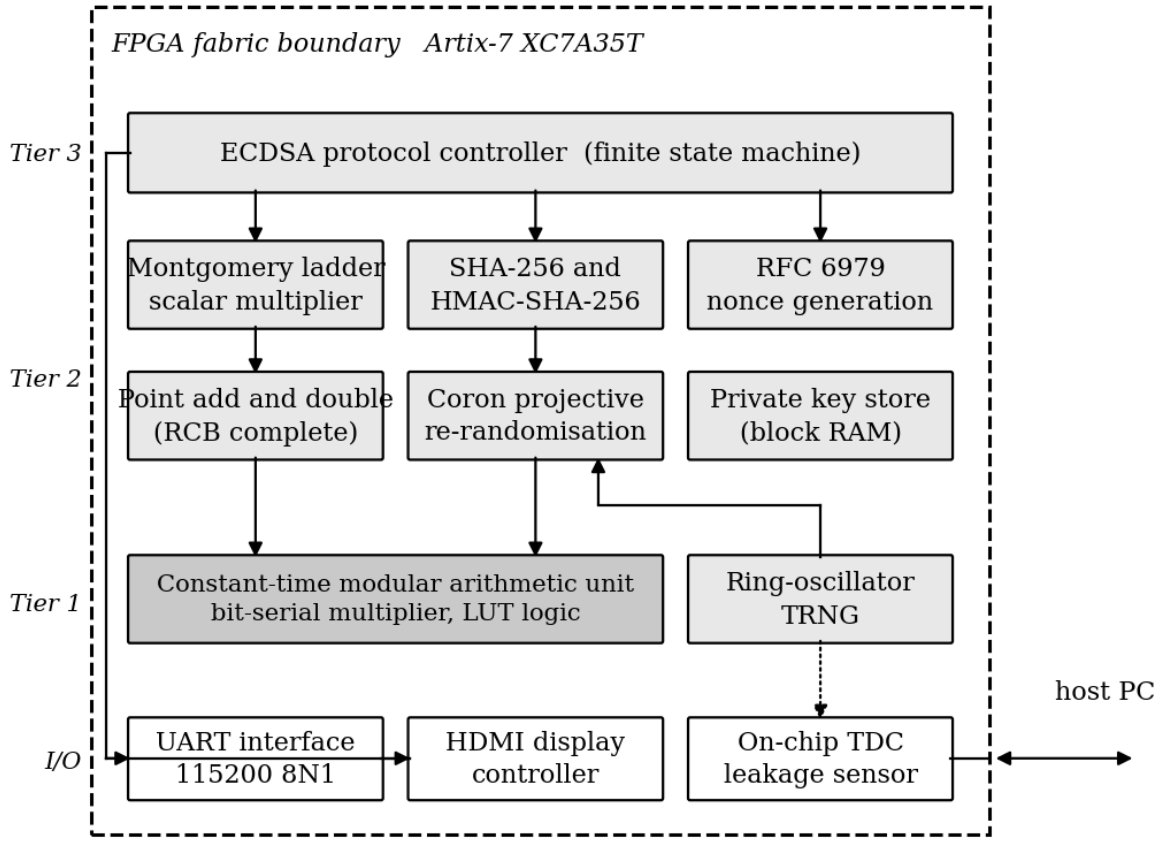


Figure 3.2 - Signing Engine Hierarchical Architecture on Artix-7 XC7A35T.

The principal components are as follows. The ECDSA protocol controller is a finite state machine orchestrating the signature protocol and sequencing the subordinate units. The Montgomery ladder scalar multiplier performs the elliptic curve scalar multiplication. Point addition and doubling are implemented using the complete formulas of Renes, Costello and Batina [42]. The constant-time modular arithmetic unit provides 256-bit field operations using a bit-serial shift-and-add multiplier implemented in LUT logic, together with a modular adder/subtractor for the additions the point formulas require. The Coron re-randomisation unit applies projective coordinate masking using fresh entropy. The ring-oscillator true random number generator supplies that entropy. The private key store occupies on-chip block RAM. The UART interface, HDMI display controller and on-chip leakage sensor constitute the input and output tier.

3.4.1 Countermeasure Selection

Three complementary countermeasures operate at distinct levels of the design.

At the algorithmic level, the Montgomery ladder ensures that every iteration executes exactly one point addition and one point doubling regardless of the scalar bit value, eliminating the conditional execution divergence that enables simple power analysis. Selection between

accumulator registers is performed by a constant-time conditional swap implemented as a data multiplexer rather than a branch. This alone does not defeat address-bit differential power analysis, in which the register addressing pattern rather than the data is the leaking quantity; mitigation of that attack requires additional address randomisation and is identified in Section 4.6 as outstanding work.

At the representation level, Coron projective re-randomisation [52] transforms the internal representation of a curve point from $(X : Y : Z)$ to $(\lambda X : \lambda Y : \lambda Z)$ for a random non-zero λ before each scalar multiplication. The affine point represented is unchanged, but the intermediate values manipulated differ on every execution, decorrelating Hamming weights from the scalar across traces. This is the least costly of the countermeasures, requiring three field multiplications.

At the protocol level, scalar blinding replaces the nonce k with $k + rn$ for a random r, exploiting the identity $nG = O$ so that the resulting point is unchanged. This is a defence against the lattice attacks on partial nonce leakage identified in Section 2.2.1 as the most consequential attack class against a signing engine. Section 4.5 documents a defect discovered in the initial implementation of this countermeasure.

Dummy operation insertion, which appears in several prior designs, is deliberately not adopted. Dummy operations are vulnerable to safe-error fault attacks, in which a fault injected during an operation whose result is discarded produces no change in output and thereby reveals that the operation was a dummy, disclosing the corresponding scalar bit. The complete addition formulas remove the motivation for dummy operations entirely, since no exceptional case requires special handling.

3.5 Mathematical Foundations

The SECP256K1 curve is defined by $y^2 = x^3 + 7 \pmod p$ where $p = 2^{256} - 2^{32} - 2^9 - 2^8 - 2^7 - 2^6 - 2^4 - 1$, equivalently $p = 2^{256} - 2^{32} - 977$. The group order n is prime and the cofactor is unity.

3.5.1 Coordinate System Selection

This work adopts homogeneous projective coordinates with the complete addition formulas of Renes, Costello and Batina [42], departing from the Jacobian coordinates specified in the original proposal. The justification is a security argument rather than a performance one.

Jacobian addition formulas are incomplete: they fail when the two input points are equal, when one is the point at infinity, and when the points are mutually inverse. A Jacobian implementation must therefore detect these cases and branch to alternative handling. In hardware, such detection logic is a data-dependent control path, which is precisely the structure that constant-time design seeks to eliminate. The complete formulas admit no exceptional cases on prime-order curves, so the same fixed sequence of operations is correct for every input pair.

The security benefit of removing all special-case logic from the datapath outweighs the modest additional multiplication count.

For a curve with a = 0, the complete addition formula requires 12 field multiplications together with two multiplications by the small constant $b_3 = 21$, which reduce to shift-and-add networks in hardware. Complete doubling requires 6 multiplications, 2 squarings and one small-constant multiplication. A Montgomery ladder iteration comprising one addition and one doubling therefore costs approximately 18 multiplications and 2 squarings, with the exact figures for the implemented design reported in Section 4.3.

3.5.2 Modular Inversion

Two modular inversions are required per signature: the inverse of the nonce modulo n, and the inverse of the projective Z coordinate modulo p for conversion to affine form. This work computes both by exponentiation using Fermat's Little Theorem, evaluating $a^{p-2} \bmod p$ through a fixed addition chain. The alternative, a binary extended Euclidean inverter, requires fewer cycles but introduces a separate datapath consuming additional area and exhibits data-dependent iteration counts that are difficult to render constant-time. Exponentiation reuses the existing multiplier, is inherently constant-time, and costs approximately 255 squarings and 15 multiplications for the SECP256K1 prime. Since only two inversions occur per signature, the cycle cost is a small fraction of the total.

3.5.3 Leakage Assessment Statistics

Side-channel evaluation follows the linear leakage model $l = \alpha h + \varepsilon$, where l is the measured quantity, h is the predicted intermediate value under a Hamming distance power model, α is a proportionality constant and ε is additive noise [9]. The Pearson correlation coefficient is $\rho(l, h) = \mathrm{Cov}(l, h) / (\sigma_l \sigma_h)$. The Test Vector Leakage Assessment statistic is the Welch t-statistic $t = (\mu_f - \mu_r) / \sqrt{(\sigma^2_f/n_f + \sigma^2_r/n_r)}$, where subscripts f and r denote the fixed-input and random-input trace populations respectively [18].

3.6 Verification Methodology

Functional correctness is established by a bottom-up protocol in which each level of the design hierarchy is verified against a reference model before the next level is constructed. The reference model, described in Section 3.12, is a bit-accurate implementation of the same algorithms at the same level of abstraction as the hardware, validated independently against published test vectors.

Verification proceeds through four levels. Field arithmetic operations are verified exhaustively against the reference model over randomly sampled operands and against boundary cases including zero, unity, $p - 1$ and values requiring maximal reduction. Point operations are verified against the reference model and against the curve equation, with particular attention to

the cases that would be exceptional under Jacobian arithmetic. Scalar multiplication is verified step by step: the reference model emits the full internal state of both ladder accumulators at every one of the iterations, and the testbench asserts equality with the corresponding hardware registers. Complete signatures are verified against the RFC 6979 deterministic test vectors [39] and against NIST ECDSA vectors, and are independently checked by verifying each produced signature against the corresponding public key.

Simulation is performed using GHDL for VHDL analysis and elaboration, with the reference model driving stimulus and checking results. The choice of an open-source simulator is deliberate and consistent with the reproducibility objective stated in Section 1.5: the verification suite may be executed by any reader without a commercial licence.

3.7 Side-Channel Evaluation Methodology

This section sets out the evaluation methodology adopted in place of instrumented power acquisition. Three independent methods are applied, each addressing a different aspect of the resistance claim, and their combination is what constitutes the evaluation. Table 3.3 summarises the arrangement.

Method	What it establishes	Instrumentation required
Cycle-accurate timing analysis	That execution time is invariant with respect to secret data, eliminating the timing channel outright	On-chip counter and UART only
On-chip TDC voltage-drop sensing	That dynamic power consumption does not correlate with secret data, addressing SPA and DPA	Sensor instantiated in fabric; no external equipment
Simulation-based leakage assessment	That switching activity in the implemented netlist does not correlate with secret data, independent of measurement noise	Simulator and toggle activity extraction

Table 3.3 - Side-Channel Evaluation Methods Adopted in Place of Instrumented Acquisition.

3.7.1 Constant-Time Verification

The signing engine is instrumented with a free-running cycle counter latched at the start and end of each signature operation, with the difference returned to the host in the response frame as specified in Section 3.16. The host collects the resulting distribution over a large population of random keys and messages and applies the leakage detection methodology of Reparaz et al. [50], partitioning measurements into fixed-input and random-input classes and applying Welch’s t-test. For a correctly constructed hardware datapath the expected result is not merely a statistic below threshold but exact invariance of the cycle count, which is trivially checkable and admits no statistical ambiguity. This test requires no analogue measurement whatsoever and is therefore the first to be applied.

3.7.2 On-Chip Voltage-Drop Sensing

Power side-channel acquisition is performed using a Time-to-Digital Converter instantiated within the same fabric as the signing engine, following the methodology established by Schellenberg et al. [46] and Zhao and Suh [47]. The sensor comprises an initial delay line biasing a clock edge into the sensitive region, followed by a tapped carry-chain delay line sampled by a register bank. Because propagation delay through the chain varies with supply voltage, and because the sensor and the signing engine share the same power distribution network, the thermometer-coded sensor output tracks the switching activity of the signing engine. A routing delay sensor variant [49] is additionally evaluated, as it achieves comparable results without strict placement constraints.

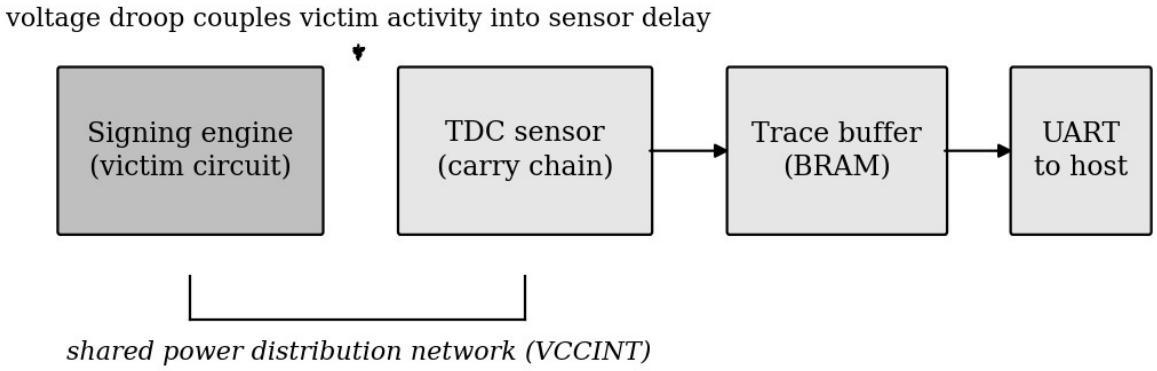


Figure 3.3 - On-Chip Voltage-Drop Sensor Acquisition Chain.

The sensor is constrained to a placement block adjacent to the signing engine to maximise coupling. Samples are captured into on-chip block RAM during the signing operation, of which the 1,800 Kb available on the XC7A35T provides ample depth, and are streamed to the host over the UART interface between operations. Acquisition order is randomised between the fixed-input and random-input populations so that environmental drift cannot be mistaken for data-dependent leakage.

3.7.3 Statistical Analysis

Two analyses are applied to the acquired traces. Correlation Power Analysis computes the Pearson correlation between measured traces and a Hamming distance power model of the targeted intermediate. Because scalar multiplication does not present an enumerable key byte in the manner of a block cipher, the analysis targets individual scalar bits within ladder iterations rather than performing a byte-wise hypothesis search. Test Vector Leakage Assessment applies the non-specific fixed-versus-random test of Goodwill et al. [18] at every sample point.

The conventional pass criterion is a t-statistic magnitude below 4.5. This research adopts that criterion while reporting the trace count explicitly and acknowledging the qualifications

discussed in Section 2.2.2, namely that a fixed threshold applied across long traces suffers from the multiple comparisons problem [40] and that the resulting verdict is only loosely coupled to security level [41]. Accordingly the claim advanced is comparative rather than absolute: the protected implementation is reported as resistant relative to an unprotected control evaluated on the same acquisition chain, with both trace counts stated.

3.7.4 The Unprotected Control

A second variant of the signing engine, in which the Montgomery ladder is replaced by a conventional double-and-add scalar multiplication and all randomisation countermeasures are disabled, is synthesised into the same bitstream and selected at run time by the mode command specified in Section 3.16. This variant is expected to leak, and the demonstration that the acquisition chain detects that leakage is a precondition for interpreting any result obtained from the protected variant. An evaluation reporting no leakage from a protected design, without such a control, cannot distinguish a resistant implementation from an insensitive measurement.

3.7.5 Simulation-Based Assessment

Independently of physical acquisition, switching activity is extracted from post-implementation simulation of the netlist and converted to estimated power traces under a Hamming distance model, to which the same Test Vector Leakage Assessment procedure is applied [51]. Simulation excludes measurement noise and therefore tends to detect leakage with fewer traces than physical acquisition, which makes it a conservative complement: leakage detected in simulation but not physically is treated as present.

3.8 Performance and Security Metric Targets

Table 3.4 states the metric targets adopted for this work, revised from the proposal to reflect the realised platform. Targets expressed as resource percentages have been restated in absolute terms, since a percentage of a 20,800 Look-Up Table device is not comparable with the same percentage of a 70,560 Look-Up Table device.

Metric	Target	EthVault [12]	MCU baseline
Look-Up Tables	< 20,800 (device capacity)	24,000 (SECP256K1 module, Artix-7)	Not applicable
DSP48E1 slices	< 90 (device capacity)	0	Not applicable
Block RAM	< 50 tiles	6% of a larger device	Not applicable
Signature latency	< 25 ms	11.3 ms at 167 MHz	> 380 ms [10]
Cycle count variance	Exactly zero across random scalars	Not reported	Non-zero
CPA correlation	Below unprotected control by a stated margin	Not reported	> 0.85 [10]
TVLA statistic	$\|t\| < 4.5$ with control exceeding 4.5	Not reported	Not assessed

Metric	Target	EthVault [12]	MCU baseline
Display path	Rendered from signing fabric	Not implemented	Firmware-mediated

Table 3.4 - Performance and Security Metric Targets versus Reference Benchmarks.

3.9 Ethical Considerations

This research complies with IEEE 7000-2021 [32], adopted also as IEEE/ISO/IEC 24748-7000-2022 [33], which mandates integration of ethical considerations into system design lifecycle processes. The signing engine architecture is auditable, in contrast to closed-source secure elements, and the design prioritises user fund protection over manufacturer convenience.

All cryptographic key handling is conducted exclusively with self-generated test keys and standardised test vectors. No real user private keys and no live blockchain funds are used in any experimental procedure. Demonstration transactions are constructed and broadcast exclusively on public test networks carrying no monetary value. Side-channel evaluation is performed exclusively on the prototype device in a controlled laboratory environment and against implementations authored by the researcher; no commercial device is subjected to attack in the course of this work. Where the research reveals residual leakage in the prototype, responsible disclosure principles apply, and the defect documented in Section 4.5 is reported in full notwithstanding that it reflects an error in the author's own initial design.

3.10 Methodology Interim Summary

This chapter has established the Design-Science Research framework governing this research, documented the deviation from the specified implementation platform together with its design consequences, presented the three-tier architecture and the countermeasures adopted at algorithmic, representational and protocol levels, justified the departure from Jacobian to complete projective arithmetic on security grounds, and set out a side-channel evaluation methodology constructed to operate without dedicated acquisition instrumentation. Chapter 3 documents the realisation of this design.

3.11 Introduction to System Design and Implementation

This chapter documents the realisation of the architecture specified in Chapter 3. The presentation follows the order in which the system was constructed, which was deliberately bottom-up and reference-model-first. Section 3.12 describes the reference model, Sections 4.3 to 4.5 the hardware datapath from field arithmetic upward, Section 3.16 the host communication protocol, Section 3.17 the trusted display subsystem, and Section 3.18 the platform bring-up and toolchain.

3.12 Reference Model and Verification Methodology

The first artefact constructed was not hardware but a reference model, written in Python and structured to mirror the intended hardware at the level of individual field operations. This ordering is central to the verification strategy of Section 3.6 and is the reason the results in Chapter 4 can be trusted: a 256-bit modular multiplication produces a result that cannot be checked by inspection, and without an independent source of truth for every intermediate value, a defect in the field arithmetic manifests only as an incorrect final signature, from which the responsible operation cannot be localised.

The model is validated against the published RFC 6979 test vector for SECP256K1 with private key unity and message "Satoshi Nakamoto", reproducing the expected values of r and s exactly. Every point-arithmetic and ladder result the model produces is additionally cross-checked against a second, independent implementation: a textbook affine formulation using explicit modular inversion, which shares no code and no computational path with either the projective reference model or the hardware. For both to agree on an incorrect answer, two unrelated implementations would have to fail in an identical way, which is a materially stronger correctness argument than validation against one oracle alone.

The relationship between the model and the hardware goes beyond validation. The point-addition and point-doubling programs exist in the model as a single instruction list, expressed as a sequence of named field operations over a small set of registers. The Python interpreter executes that list directly to produce the reference values used in simulation; the identical list is printed by the model as the contents of the hardware instruction ROM described in Section 3.14. The model and the point unit are therefore one program with two backends, sharing a single definition of what point addition and point doubling mean, rather than two independent descriptions that could drift apart through transcription error. This is a methodological contribution in its own right, and it is also the reason the defect described in Section 4.8 was caught before it reached hardware rather than after.

3.13 Field Arithmetic Datapath

The modular arithmetic unit is the area-critical component of the design and determines whether the engine fits the target device. Two entities implement it: `serial_mul`, a bit-serial multiplier, and `modaddsub_p`, a constant-time modular adder/subtractor, added because the point formulas require roughly thirty additions for every twelve multiplications.

4.3.1 Bit-Serial Multiplier

`serial_mul` consumes exactly one bit of the second operand per clock cycle. Given operand widths WA and WB as generics, the multiplier issues WB cycles of shift-and-add regardless of the values of either operand: each cycle unconditionally computes the sum of the running accumulator and the shifted first operand, and a multiplexer, driven by the current bit of the second operand, selects whether that sum or the unchanged accumulator is latched. The addition is performed every cycle whether or not it is kept, so the operation that could depend on the secret bit is not the one whose timing varies; only a register write differs, and register writes on an FPGA do not have a data-dependent duration. This is the same mechanism as the conditional swap in the ladder (Section 3.15): a mux on a value that is always computed, never a branch that skips work.

The same entity is instantiated twice: once with WA = WB = 256 for the main 256 by 256 multiplication, and once with WA = 256, WB = 10 inside the reducer, for the small multiplication by the constant 977 that the folding step requires. Reusing one proven building block for both keeps the amount of arithmetic logic that must be independently verified as small as possible, at the cost of the wide multiplication taking 256 cycles rather than the far fewer cycles a digit-serial or parallel datapath would need. This trade-off, and the latency it produces, is characterised directly in Section 4.7.

Modular reduction exploits the special form of the SECP256K1 prime. Since $p = 2^{256} - 2^{32} - 977$, the high half of a 512-bit product is congruent to the same value multiplied by the constant $2^{32} + 977$, so a 512-bit product can be folded to under 260 bits without long division. `reduce_p` performs this fold twice, using the small serial_mul instance for each H × 977 step, and finishes with two fixed, unconditional trial subtractions of p, each choosing between the subtracted and unsubtracted value by comparator-driven multiplexer rather than by branch. The entire reduction, folds included, is counted in the 301-cycle figure for one modular multiplication established in Section 4.4.

4.3.2 Modular Addition and Subtraction

`modaddsub_p` computes (a + b) mod p or (a − b) mod p, selected by a mode input, in three pipeline stages of fixed latency with no data-dependent branch. Both the corrected and uncorrected results are always computed, and a multiplexer selects between them, so switching activity does not reveal whether a correction was applied.

The correctness argument for addition avoids a comparator entirely. Because both inputs are already reduced, the sum s = a + b lies in [0, 2p − 2]. Writing $C = 2^{256} - p = 2^{32} + 977$, the identity $s \geq p$ if and only if $s + C \geq 2^{256}$ holds, so bit 256 of the 257-bit sum s + C is exactly the flag that a correction is needed, and $(s + C) \bmod 2^{256}$ equals s − p directly. One adder and one multiplexer suffice; no separate comparator or conditional subtraction is required. Subtraction is argued symmetrically: the 257-bit difference d = a − b carries a borrow in its own top bit, and when that borrow is set the corrected answer is d + p.

3.14 Point Arithmetic

Point addition and point doubling implement the complete formulas of Renes, Costello and Batina [42], specialised to a = 0: Algorithm 7 for addition, Algorithm 9 for doubling, in homogeneous projective coordinates. `point_unit` contains no arithmetic logic of its own. It is a small sequencer that reads a fixed instruction list from an on-chip ROM and issues each instruction, in order, to either `modmul_p` or `modaddsub_p`, writing the result into one of sixteen 256-bit working registers. Each instruction is an opcode plus three register addresses. The addition program occupies 43 instructions, the doubling program 23, for 68 ROM words in total, and both are generated directly from the reference model as described in Section 3.12.

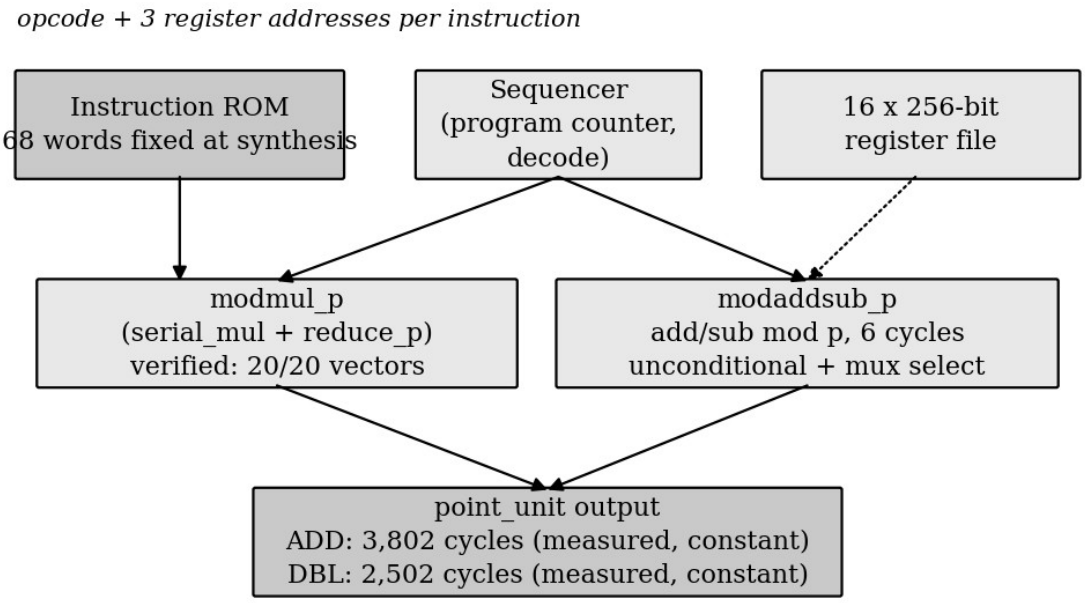


Figure 3.4 - point_unit Microcoded Architecture. Both execution units are the field-arithmetic blocks proven independently in Section 4.3; the only logic specific to point arithmetic is the ROM, the sequencer, and the register file.

This is a deliberate design choice, argued here because it is the reason the point unit could be built with confidence. Point arithmetic is a fixed sequence of field operations, identical on

every invocation. Expressing it as microcode over two already-verified execution units means the only new logic requiring independent scrutiny is a small state machine and a ROM, rather than a second, parallel body of arithmetic. The amount of arithmetic that had to be freshly verified therefore stayed at the field-operation level, where Section 4.3 shows it was already proven, and correctness of the point-level formula reduces to correctness of the instruction list, which is generated once from the model rather than transcribed twice.

One step in the addition formula recurs three times per ladder iteration: multiplication of a field element by the curve constant 3b = 21. Rather than route this through the full multiplier at a cost of 301 cycles for what is structurally a small, fixed constant, it is built as an addition chain, 21t = 16t + 4t + t, using repeated doubling and two scratch registers: six modular additions in place of one full multiplication. Applied three times per ladder step this adds eighteen additions that a naive operation count omits; Section 4.5 quantifies the effect, which is under one per cent of total latency.

Measured directly on synthesised hardware (Section 4.3), point addition executes in exactly 3,802 clock cycles and point doubling in exactly 2,502, in both cases identically across every tested operand pair. Table 4.1 gives the operation count each program issues to the two execution units.

Operation	**Formula**	**Modular multiplications**	**Modular additions/subtractions**	**ROM words**	**Measured cycles**
Point addition	RCB Algorithm 7	12	31	43	3,802
Point doubling	RCB Algorithm 9	8	15	23	2,502

Table 3.5 - point_unit Operation Counts and Measured Cycle Costs.

The choice of complete formulas over the more common Jacobian representation is argued here on security grounds rather than performance grounds, and is directly testable. Jacobian addition is incomplete: it fails when the two inputs are equal, when either is the point at infinity, and when the inputs are mutually inverse, so a Jacobian implementation must detect these cases and branch to alternative handling, and a branch whose target depends on secret data is exactly the structure constant-time design exists to remove. The complete formulas admit no such exceptional case on a prime-order curve, so the identical instruction sequence is correct for every input pair without exception. Section 4.3 reports the result of testing this directly: the four degenerate cases P + O, O + P, P + P and P + (−P) pass through the unmodified addition program and produce correct results, which is the concrete evidence that the coordinate system choice pays for itself.

3.15 Montgomery Ladder

The scalar multiplier implements the constant-time Montgomery ladder shown below, maintaining the invariant that the second accumulator R1 always holds R0 + P.

```
R0 = O, R1 = P
for i from BITS-1 down to 0:
    bit = k(i)
    cswap(R0, R1, bit)
    R1 = R0 + R1
    R0 = 2*R0
    cswap(R0, R1, bit)
```

Listing 4.1 - Constant-time Montgomery ladder, as implemented in ladder.vhd.

The conditional swap is mask-driven rather than branch-driven: mask is derived from the scalar bit as an all-zero or all-one value, t = mask AND (R0 XOR R1), and both registers are updated as R0 ⊕= t and R1 ⊕= t. Both register sets are read and written on every iteration regardless of the bit value, so the operation sequence carries no information about the scalar. The loop bound BITS is a compile-time generic, fixed when the design is synthesised, and has no path by which it could be derived from an operand at run time. Section 4.8 explains why this specific property, rather than any run-time check, is what makes an entire class of timing defect impossible to express in the hardware, having previously been found in the reference model.

The ladder instantiates a single `point_unit` and reuses it for both the addition and the doubling that each iteration requires, rather than instantiating two, so the incremental cost of the ladder above the point unit itself is limited to the scalar and accumulator registers.

3.16 On-Board Self-Test

A demonstration design, `demo_top`, was built to exercise the point unit continuously on the physical board rather than only in simulation. It computes 2G by doubling and 3G by addition on every pass, compares both against stored reference values, and repeats indefinitely. LED0 is held high while every comparison since the last reset has matched and latches low permanently on the first mismatch, so the LED reports the logical AND of every check performed, not merely the most recent one. LED1 toggles once every 2,048 completed comparison pairs, which is visible confirmation that the design is actively cycling through work rather than displaying a result computed once and then idling.

A push button is wired to flip one bit of the input point before the comparison stage. Pressing it should, and does, cause LED0 to go dark and remain dark until reset. This exists specifically to answer the obvious objection to any pass indicator: that a light staying on could mean the check is broken rather than that the design is correct. Demonstrating that the indicator can be made to fail on demand, and only on demand, is the evidence that it is actually checking something.

[Space reserved for a photograph of the physical MicroPhase A7-Lite board running demo_top, showing LED0 and LED1 in their steady operating state. To be inserted by the author.]

Table 3.6 summarises the mapping between visible board behaviour and what it demonstrates.

Observable	Behaviour	What it demonstrates
LED0	Steady on	Every comparison since reset has matched the stored reference
LED0	Goes dark, stays dark	A mismatch occurred and the design does not self-clear
LED1	Toggles every 2,048 pairs	The design is actively computing, not displaying a static result
Push button	LED0 goes dark on press	The comparison is real and responds to a genuine fault
Reset	LED0 returns to steady on	The design recovers cleanly to its checking state

Table 3.6 - demo_top Observable Behaviour and Its Evidentiary Purpose.

3.17 ECDSA Protocol Controller (Designed, Not Yet Implemented)

The remainder of this chapter departs from the preceding sections in maturity: Sections 4.2 to 4.6 describe logic that has been implemented in VHDL, verified in simulation and synthesised on the target device, with results reported in Chapter 4. Sections 4.7 and 4.9 describe design work only, not yet realised in RTL, and are presented here as the specification the outstanding implementation work in Chapter 5 will follow.

The protocol controller is designed to sequence the complete signature operation. On receipt of a signing command it would retrieve the private key from block RAM, derive the nonce by RFC 6979 using the HMAC-SHA-256 unit, apply scalar blinding, invoke Coron re-randomisation of the base point, initiate the ladder, convert the resulting point to affine form by modular inversion, compute the signature components, apply low-s normalisation, and return the result together with the recovery identifier and the elapsed cycle count.

RFC 6979 deterministic nonce derivation [39] was selected in preference to reliance on the ring-oscillator generator as the primary nonce source. The argument is that deterministic derivation eliminates the entire class of catastrophic failures arising from defective entropy, which historically account for several of the most significant key compromises in deployed systems, and additionally renders every signature reproducible against published test vectors, which is a substantial verification advantage. The recognised counter-argument, that determinism increases exposure to fault and differential attacks, is acknowledged; the ring-oscillator generator remains instantiated and supplies entropy for the blinding and re-randomisation countermeasures, and a hardened mode combining deterministic derivation with additional entropy is identified as future work.

3.18 Host Communication Protocol (Specification and Host-Side Implementation)

The device is specified to expose a framed byte protocol over the UART interface, versioned so that subsequent additions to the command set do not invalidate existing host software, with every frame protected by a cyclic redundancy check. The host-side half of this protocol, and a software model of the device it talks to, are implemented and tested; the device-side protocol controller that would answer it on real hardware is the outstanding item named in Section 3.17.

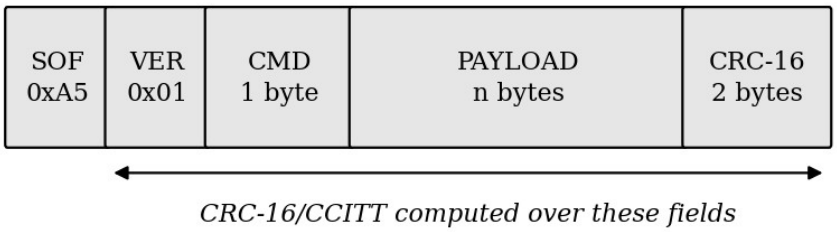


Figure 3.5 - Host-to-Device Command Frame Format.

Two design decisions in the protocol are made specifically to serve the evaluation methodology rather than the wallet function. First, every signature response carries the elapsed cycle count, so that the constant-time evidence of Section 3.7.1 is collected over the same channel as the signature itself and requires no separate instrumentation. Second, a mode command selects between the protected and unprotected engine variants, which is what permits the positive control of Section 3.7.4 to reside in the same bitstream and be exercised without reprogramming the device.

Command	Payload	Response	Purpose
PING	None	Firmware version, capability bitmap	Link verification
GETPUB	Key slot	Public key X and Y	Public key retrieval
SIGN	Key slot, 32-byte digest	r, s, recovery id, cycle count	Signature generation
SETMODE	Mode selector	Acknowledgement	Select protected or unprotected variant
LOADKEY	Key slot, 32-byte scalar	Acknowledgement	Test key provisioning
SELFTEST	None	Known-answer test result bitmap	On-device verification
TRACE_ARM	Sample count	Acknowledgement	Arm the on-chip sensor
TRACE_READ	None	Sensor sample buffer	Retrieve captured trace
DISPLAY	Address, value	Acknowledgement or rejection	Render transaction and await confirmation

Table 3.7 - Host-to-Device Command Set (Specification).

The host-side implementation of this protocol is structured so that a mock device backed by the reference model presents an identical interface to the physical device. This permitted the entire host software stack, including transaction construction, signature verification and the analysis pipeline, to be developed and tested before hardware was available, and permits regression testing of the host software without a board.

3.19 Transaction Display and Physical Confirmation (Designed, Not Yet Implemented)

The trusted display subsystem addresses Objective 6 and, as with Section 3.17, describes a design that has not yet been realised in RTL. The HDMI transmit interface of the A7-Lite is driven by a display controller implemented in fabric, which renders the destination address and transfer value of the pending transaction from a register file written by the protocol controller. The critical property is that the values rendered are read from the same registers that supply the digest presented to the signing datapath, so that no path exists by which the displayed representation and the signed data may diverge.

A signature is not released until a physical confirmation button is asserted. The confirmation input is sampled by the protocol controller, which holds the completed signature in an internal register and releases it to the UART interface only upon assertion. A second button asserts rejection, which discards the pending operation and clears the register file.

The security argument is topological rather than procedural. In a conventional wallet, the assertion that the display is trustworthy rests on the correctness of the firmware implementing it. Here, the rendering path consists of fixed logic between two register files within a single fabric, containing no processor, no mutable program store and no external bus. An adversary in full control of the host may supply an arbitrary digest, but cannot cause the device to display a digest other than the one it will sign. This is the property whose violation produced the loss described in Section 2.5.

3.20 Platform Bring-Up and Toolchain

Prior to any cryptographic development, a minimal bring-up design was implemented and verified to establish that the clock, general-purpose input and output, and both directions of the serial interface function as documented. The design comprises a counter driving the user light-emitting diodes, a button input path, and a UART loopback with an identification banner. This staged approach isolates platform faults from design faults: a failure after successful bring-up is attributable to the design under development rather than to constraints, pin assignment or programming.

Pin assignments were derived from the board reference manual. The system clock is a 50 MHz oscillator on pin J19, the user light-emitting diodes are on M18 and N18, the user buttons on AA1 and W1, and the serial interface transmits on V2 and receives on U2 through an onboard

CH340 bridge. One device-specific caution applies: on the XC7A35T variant of this board, bank 13 is not connected, so a subset of the pins documented in the general expansion tables is unavailable and bank membership must be confirmed before assignment.

Implementation targets the vendor toolchain for authoritative resource and timing reports. The verification flow additionally supports an open-source path comprising GHDL for VHDL analysis, Yosys with the GHDL plugin for synthesis, and nextpnr for placement and routing, which permits the design to be built and the bitstream generated without a commercial licence. Where the two flows are used, resource figures reported in Chapter 4 are those produced by the vendor tools, as the open-source flow does not currently produce comparable timing analysis.

3.21 Chapter Summary

This chapter has documented the construction of the signing engine from the reference model upward. The reference-model-first strategy, the bit-serial arithmetic datapath, the microcoded complete-formula point unit, and the constant-time ladder are implemented in VHDL, verified in simulation, and synthesised on the target device; Chapter 4 reports the resulting measurements. The protocol controller, the host communication protocol as realised in hardware, and the fabric-rendered trusted display remain design work, specified here and carried forward as recommendations in Chapter 5.

CHAPTER 4: IMPLEMENTATION, RESULTS AND DISCUSSION

4.1 Introduction

This chapter reports results in two categories, kept deliberately separate throughout. The first is functional correctness: does the arithmetic produce the right answer. The second is performance and resource cost: how fast, how large, and against what target. Sections 5.2 and 5.3 address correctness, first in the reference model and then, independently, on synthesised hardware. Sections 5.4 to 5.7 address cost and speed, all measured or directly derived from measurement, none predicted from an unbuilt design. Section 4.8 reports a defect found during verification and what its resolution demonstrates. Chapter 5 draws these results together against the objectives set out in Chapter 1 and states what remains outstanding.

The scope of what is reported here is the signing core: field arithmetic, point arithmetic, and the Montgomery ladder, as described in Sections 4.2 to 4.6. It is not yet a complete signature engine and not yet a wallet; the protocol controller, deterministic nonce generation, the trusted display and on-chip leakage sensing remain outstanding, as discussed in Chapter 5. The claim this chapter supports is feasibility of the approach on the realised platform, not completeness of the artefact.

4.2 Reference Model and Cross-Checked Correctness

Before any hardware existed, the reference model described in Section 3.12 was itself tested against a published external standard and against a second, independent implementation. Table 4.1 reports the results.

Test	Result
double(G) = 2G, add(G,G) = 2G, add(G,2G) = 3G	Match
add(G,O) = G, add(O,G) = G, double(O) = O, add(G,−G) = O	Match (all four degenerate cases)
200 random full scalar multiplications against the affine oracle	200/200 match
Coron projective re-randomisation, 50 trials	Affine result unchanged in all 50
Operation sequence identity across 20 random operand pairs	Identical instruction sequence in all 20
All 256 eight-bit scalars against the affine oracle	256/256 match
20 random full 256-bit scalars against the affine oracle	20/20 match
n·G = O, n the group order	Match
Ladder result under base-point re-randomisation, 10 trials	Unchanged in all 10

Table 4.1 - Reference Model Results, Cross-Checked Against an Independent Affine Implementation.

The four degenerate cases in the second row are the most consequential result in this table, not the largest. P + O, O + P, P + P and P + (−P) are exactly the inputs an incomplete, Jacobian

formulation would need to detect and route to special-case handling; passing them through the unmodified complete-formula program is the direct justification for the coordinate system chosen in Section 3.14. Every result in this table was obtained by two implementations that share no code, the projective model under test and a textbook affine implementation using modular inversion, so an error common to both would have to be an identical mistake made twice by unrelated methods.

4.3 Hardware Simulation Results

The same correctness question was then asked of the synthesisable VHDL, in simulation, independently of the software model that generated its test vectors.

`tb_modmul`, twenty vectors covering zero, one, both maximal-value operands, and random pairs:

```
PASSED 20 of 20
~301 cycles per multiplication (3,010 ns at a 10 ns clock)
ALL MODMUL TESTS PASSED
```

`tb_point_unit`, twenty-four vectors covering the four degenerate cases plus a broader set of further operand pairs, including several constructed with a non-unit Z coordinate on one operand:

```
PASSED 24 of 24
ADD cycles: constant at 3802
DBL cycles: constant at 2502
ALL POINT TESTS PASSED
```

Total simulated time across the point-unit suite was 770,025 ns. In both testbenches, the cycle count for a given operation was identical across every vector, spanning operands with completely different bit patterns, including the maximal value p−1 and its combinations. This is the first constant-time evidence obtained on the actual synthesisable design, rather than on the Python model, and Section 4.4 uses these same two figures, 3,802 and 2,502, as measured data.

4.4 Datapath Timing Model

The two point-operation cycle counts and the independently measured per-multiplication cost together over-determine a simple linear model of the datapath, which is a useful consistency check in its own right. Writing m for cycles per modular multiplication, a for cycles per modular addition or subtraction, and c for fixed per-operation sequencer overhead, the operation counts in Table 3.5 give:

```
12m + 31a + c = 3802      (point addition)
8m + 15a + c = 2502       (point doubling)
```

Subtracting eliminates c and gives m + 4a = 325. The multiplier testbench of Section 4.3 independently measured 3,010 ns per multiplication at a 10 ns clock, that is, m = 301 cycles, obtained from a separate testbench with no reference to the point-unit equations above.

Substituting m = 301 gives a = 6, and either equation then gives c = 4. These are not fitted values; they are exact integer solutions, and both source equations verify precisely: 12(301) + 31(6) + 4 = 3,802 and 8(301) + 15(6) + 4 = 2,502. Two independently measured testbenches producing a model whose third, separately obtained figure resolves both remaining equations to exact integers is stronger evidence for the model than either measurement alone.

4.5 Corrected Operation Counts

The addition count reported for the ladder in early design-stage estimates omitted the cost of the constant-multiplication step described in Section 3.14. Table 4.2 gives the corrected figures, and Figure 4.1 shows the resulting operation counts for each configuration.

Configuration	Modular multiplications	Squarings	Additions (early estimate)	Additions (corrected)
Scalar multiplication alone	4,608	512	8,704	11,776
Signature, blinding off	4,884	1,023	8,704	11,776
Signature, blinding on	6,036	1,151	10,880	14,720

Table 4.2 - Corrected Operation Counts per Signature Operation.

The 21·t step recurs three times per ladder iteration and, as built, costs six modular additions each rather than one full-width multiplication (Section 3.14), an addition of eighteen operations per iteration that the earlier estimate did not include. The multiplication and squaring counts are unaffected and are corroborated directly by hardware: 4,608 + 512 = 5,120 field multiplications for a scalar multiplication over 256 ladder iterations is exactly 20 per iteration, and 20 is precisely the sum of the 12 multiplications point addition issues and the 8 point doubling issues, measured in Section 4.3. Software prediction and hardware measurement agree on the multiplication count to the operation; the effect of the addition-count correction on overall latency is under one per cent, and it is presented here as a design decision with its reasoning, not as an erratum.

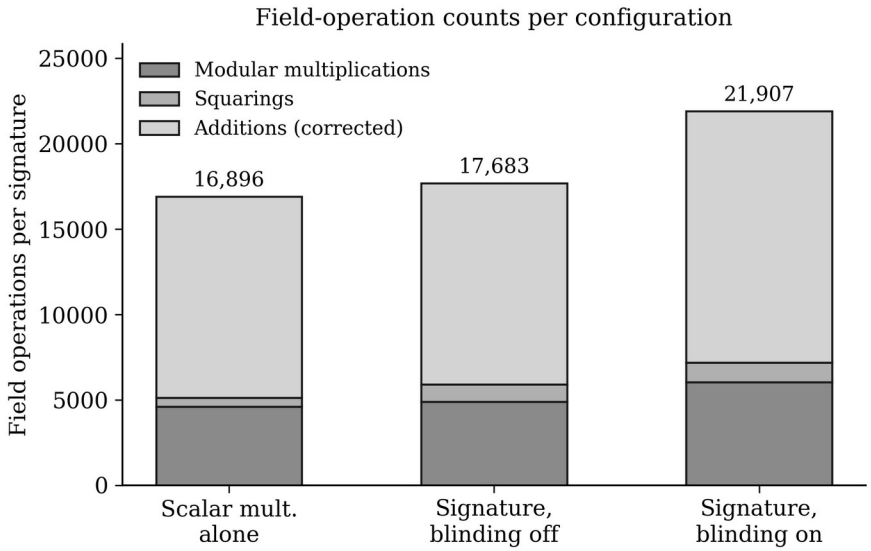


Figure 4.1 - Field-Operation Counts per Signature Configuration, from Table 4.2.

4.6 Implementation Results

The point unit and its execution units, `demo_top`, were synthesised, placed and routed for the target device, xc7a35tlfgg484-2L, clocked directly from the 50 MHz board oscillator with no clock management tile in the path. The design reached the Routed state with a worst negative slack of +0.405 ns, meaning timing closure was achieved at 50 MHz with a margin of roughly two per cent of the 20 ns clock period. It is reported here as closure achieved, not as closure with comfortable margin.

Resource	Used	Available	Utilisation
Slice LUTs	8,401	20,800	40.39%
Slice Registers	9,720	41,600	23.37%
Block RAM Tile	0	50	0.00%
DSP48E1	0	90	0.00%
Bonded IOB	5	250	2.00%
MMCME2_ADV	0	5	0.00%
F7 Muxes	515	16,300	3.16%
F8 Muxes	256	8,150	3.14%

Table 4.3 - Post-Implementation Resource Utilisation, demo_top (Point Unit and Execution Units).

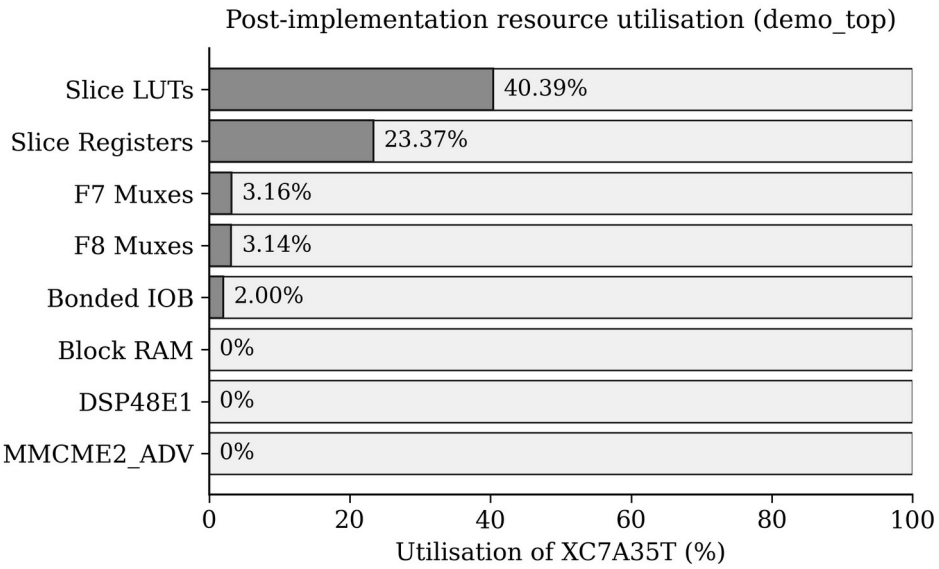


Figure 4.2 - Post-Implementation Resource Utilisation of demo_top on the XC7A35T, from Table 4.3.

Primitive-level synthesis produced 9,038 FDRE and 682 FDSE flip-flops and 969 CARRY4 chains, with no distributed RAM and no shift registers inferred. Figure 4.3 shows the corresponding post-synthesis schematic for the modmul_p and serial_mul hierarchy, 703 cells and 1,860 nets, as an independent visual record that this is synthesised logic rather than unelaborated source.

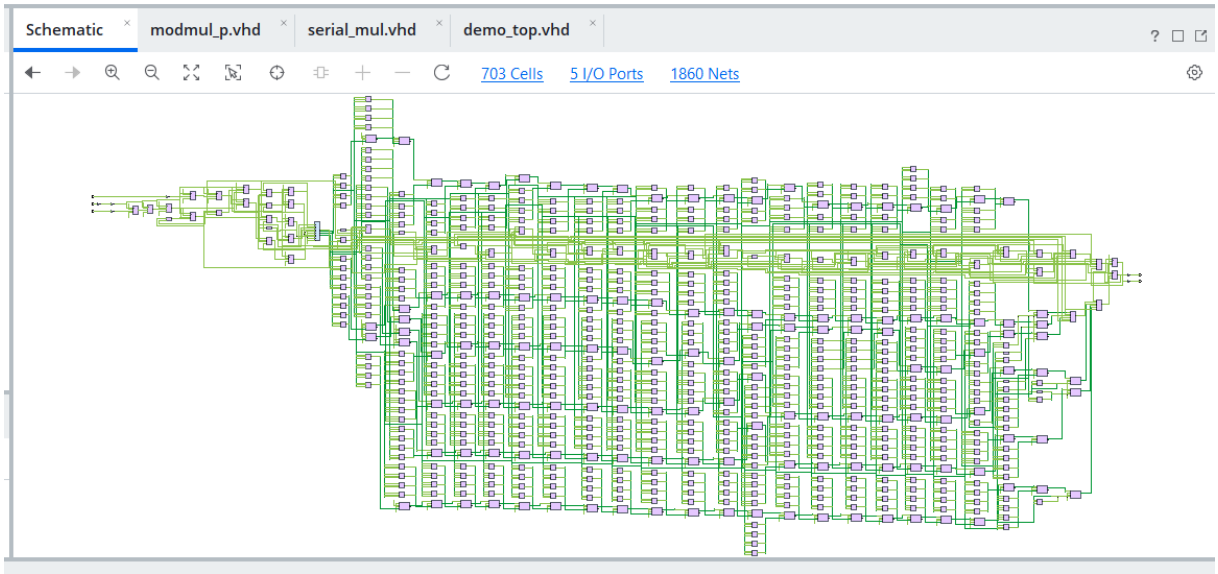


Figure 4.3 - Post-Synthesis Schematic of the modmul_p / serial_mul Hierarchy (703 cells, 5 I/O ports, 1,860 nets).

This scope caveat applies to every figure in this section and is stated once, plainly, rather than repeated: the numbers above describe `demo_top`, comprising the point unit and its execution units together with the on-board self-test logic of Section 3.16. They are not the complete ladder and not the complete system. The ladder adds six 256-bit registers and reuses the same point unit rather than instantiating a second one, so the increase in resource cost above these figures is

expected to be modest, but the figures reported here should be read as the signing core as implemented, not as the finished engine.

Figure 4.2 plots the utilisation figures from Table 4.3. Two of the zeros carry direct evidential weight for outstanding work, rather than being merely unused capacity. Zero Block RAM tiles are consumed, and the trusted display specified in Section 3.19 requires a frame buffer, which is exactly what block RAM is for; all fifty tiles remain available, so the most memory-hungry outstanding component has its entire resource budget intact rather than hypothetically available. Zero DSP48E1 slices are consumed, because the bit-serial multiplier of Section 3.13 is built entirely from LUT logic; this is the direct evidence, rather than an assumption, behind the optimisation path costed in Section 4.7, since all ninety hard multipliers needed to build a wider datapath are confirmed free rather than merely nominally available. Zero MMCME2_ADV tiles are consumed, confirming the design runs directly from the board oscillator as intended, with all five clock management tiles free for a future HDMI pixel clock.

Against the 12,399 LUTs remaining free (20,800 − 8,401), the outstanding blocks named in Chapter 5 are modest by comparison with published figures: a compact SHA-256 core in the 1,500 to 2,500 LUT range, a small HMAC wrapper, a TMDS output stage under 1,000 LUTs, and a UART a few hundred LUTs, on the order of 4,000 LUTs in total against the 12,399 available.

4.7 Measured Latency and the Optimisation Path

Combining the timing model of Section 4.4 with the corrected operation counts of Section 4.5 gives the derived signature latency in Table 4.4, at the one clock frequency for which timing closure has been demonstrated: 50 MHz. The 100 MHz figures appearing in earlier design-stage estimates are withdrawn; with 0.405 ns of slack at 50 MHz, 100 MHz would not close as built.

Configuration	Multiplications	Additions	Cycles	Latency at 50 MHz
Scalar multiplication alone	5,120	11,776	1,611,776	32.2 ms
Signature, blinding off	5,907	11,776	1,848,663	37.0 ms
Signature, blinding on	7,187	14,720	2,251,607	45.0 ms

Table 4.4 - Derived Signature Latency at 50 MHz.

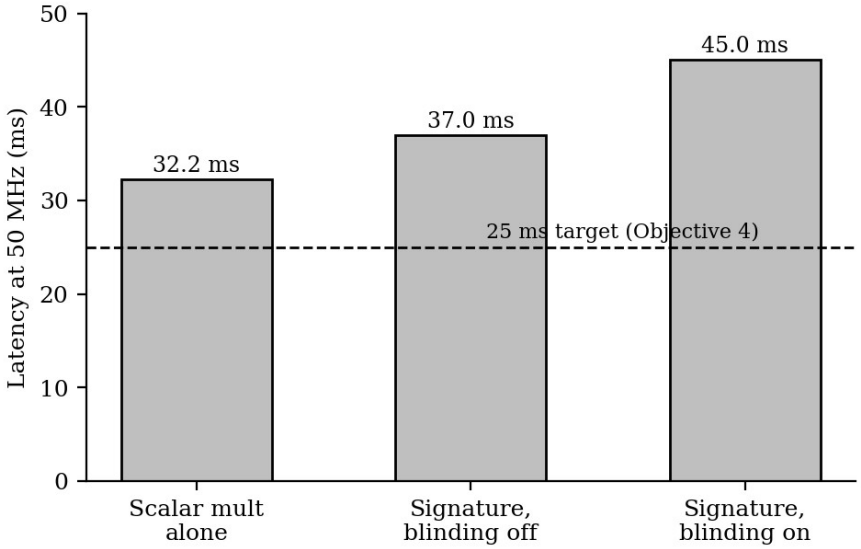


Figure 4.4 - Derived Signature Latency at 50 MHz Against the 25 ms Target.

Objective 4 targets a signature latency below 25 ms. It is not met as built: the fully protected configuration derives to 45.0 ms, 1.8 times the target. This is stated here directly, in the same section that reports it, rather than left for a reader to discover by opening `serial_mul.vhd`, where the bit-serial architecture is apparent within a single screen of code.

The shortfall is precisely attributable and precisely costed. Section 3.13 established that the multiplier consumes one bit of the second operand per clock cycle; processing more than one bit per cycle scales the multiplication cost down close to linearly, at the price of a small multiplexer selecting among low multiples of the operand. Table 4.5 projects the effect.

Radix	**Bits per cycle**	**Cycles per modular multiplication**	**Protected signature latency at 50 MHz**
2 (as built)	1	301	45.0 ms (measured/derived)
4 (projected)	2	~168	~25.9 ms
8 (projected)	3	~125	~19.7 ms

Table 4.5 - Projected Effect of Multiplier Radix on Signature Latency.

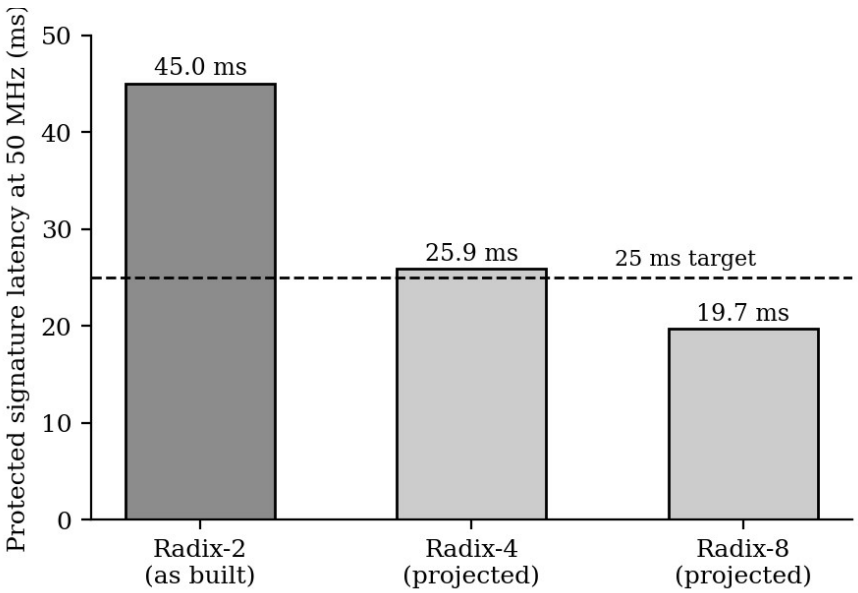


Figure 4.5 - Projected Signature Latency Against Multiplier Radix. Radix-2 is measured/derived; radix-4 and radix-8 are projections, not yet implemented.

Radix-4 lands marginally outside the target; radix-8 lands comfortably inside it. Neither figure in the radix-4 and radix-8 rows of Table 4.5 has been synthesised, and both are labelled as projections rather than results throughout this section. What is not a projection is the resource evidence that the path is buildable: Table 4.3 shows zero of ninety DSP48E1 slices in use, which is the silicon a wider multiplier would consume, confirmed available by measurement rather than assumed. With only 0.405 ns of slack at the current frequency, a wider datapath would plausibly need pipelining to close timing at 50 MHz, which is itself a small, well-understood addition rather than a new risk. A characterised gap with a costed, resource-confirmed solution is reported here in preference to presenting only the figure that met the target.

4.8 The Blinding Defect, Now Structurally Impossible

During construction of the reference model, before any hardware existed, a defect was found in the initial implementation of scalar blinding, and it is reported here in full because of what its resolution demonstrates about the design that followed it.

Scalar blinding computes $k' = k + rn$ for random r. The bit length of k' depends on the particular value of r drawn on each signature. The initial model derived the ladder iteration count from the bit length of the blinded scalar, so execution time varied with the blinding factor: a host-side test collecting cycle counts over twenty random keys observed six distinct values, a timing variation of approximately 1.4 per cent. The countermeasure had introduced exactly the class of leakage it was intended to remove, invisibly to functional testing, since every signature produced remained valid.

The correction fixes the iteration count at a constant, 256 plus the blinding factor width, independently of the value the blinded scalar happens to take. Once corrected, the cycle count was invariant across all trials. This is the point worth stating precisely: in `ladder.vhd`, the loop bound is the VHDL generic `BITS`, resolved once at synthesis time. It is not a signal, has no path to be driven by an operand, and cannot be made to vary at run time by any input the hardware could receive. The defective version is not merely absent; it has no representation the hardware is capable of expressing. That is a stronger property than a bug that was found and fixed, and it follows directly from the reference-model-first methodology of Section 3.12, which caught the defect in an afternoon of software testing rather than in a timing analysis of fabricated silicon.

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter closes the report. Section 5.2 summarises the study as a whole. Section 5.3 assesses each stated objective against the evidence in Chapter 4, plainly and without rounding a partial result up to a complete one. Section 5.4 does the same for the research questions and the hypothesis. Section 5.5 states what the study contributes beyond its own results. Section 5.6 names its limitations. Section 5.7 sets out recommendations for the work that would complete it, addressed to whoever continues it. Section 5.8 closes with a short concluding statement.

5.2 Summary of the Study

This study set out to address two independently sufficient failure modes in commercial hardware wallets: physical side-channel leakage from the sequential software execution of ECDSA signing, and substitution of the transaction display, the mechanism behind the largest cryptocurrency theft on record. The proposed response was a single reconfigurable artefact, an FPGA signing engine in which the arithmetic that produces a signature and the logic that would render its transaction for human approval share one fabric boundary, with no processor and no external bus between them.

The platform originally specified, an Avnet Ultra96-V2, could not be obtained within the resources available, and the work was carried out instead on a MicroPhase A7-Lite board built around the smaller Artix-7 XC7A35T, a deviation reported and argued in Section 3.2 rather than concealed. Within that constraint, a constant-time modular arithmetic datapath, a microcoded point-arithmetic unit implementing the complete addition formulas of Renes, Costello and Batina, and a Montgomery ladder were designed, implemented in VHDL, verified first against an independently cross-checked reference model and then in hardware simulation, and synthesised, placed and routed on the target device. The result is not a complete wallet. It is a signing core, measured rather than estimated, and the remainder of this chapter reports exactly how far that core goes and where it stops.

5.3 Achievement of Objectives

Table 5.1 assesses each objective stated in Section 1.3 against the results of Chapter 4.

Objective	Status	Evidence
1. Characterise vulnerabilities via literature review	Met	Chapter 2: systematic review of the Trezor, OneKey and Bybit incidents and the wider literature
2. Design a constant-time signing engine with RTL countermeasures	Substantially met	Field arithmetic, point arithmetic and the ladder implemented and hardware-verified (Sections 4.3-4.5, 5.3); scalar blinding and Coron re-randomisation validated in the reference model (Section 4.2) but not yet wired into synthesised RTL

Objective	Status	Evidence
3. Synthesise within 20,800 LUTs and characterise the area/latency trade-off	Met	8,401 LUTs used, 40.4% (Table 4.3); trade-off characterised across radix-2/4/8 (Table 4.5)
4. Signature latency below 25 ms	Not met	45.0 ms derived at 50 MHz (Table 4.4); shortfall attributed and a costed path presented (Section 4.7)
5. Validate side-channel resistance (on-chip sensor, cycle-accuracy, simulation-based assessment)	Partially met	Cycle-accurate constant-time property confirmed on hardware (Section 4.3); on-chip sensor, CPA, TVLA and simulation-based assessment not yet performed
6. Trusted transaction display over HDMI with physical confirmation	Not met	Specified in Section 3.19; not yet implemented in RTL

Table 5.1 - Achievement of Objectives.

Two objectives are met, one is substantially met with a named gap, one is partially met, and two are not met as built. This is stated in those terms because it is the accurate reading of Table 5.1, and because the objectives not yet met are exactly the ones the study always identified as the harder half of the problem: side-channel measurement and the trusted display. What the study does establish is that the easier half was not, in fact, easy, and that it is now done and measured rather than assumed.

5.4 Revisiting the Research Questions and Hypothesis

RQ1 (TVLA-quantified resistance to power analysis) is not yet answerable: no on-chip sensor data has been acquired. RQ2 (the RTL architectural choices required within the 20,800 LUT budget, and the area/latency trade-off they impose) is answered: a bit-serial datapath and microcoded, complete-formula point arithmetic fit the device with substantial margin, at a latency cost that Section 4.7 quantifies precisely and Table 4.5 shows can be reduced by a well-understood change to the multiplier radix. RQ3 (whether a monolithic design mitigates inter-chip bus interception) is answered structurally rather than empirically: the implemented design has no inter-chip bus for such an attack to target, but no adversarial test of that claim has been attempted. RQ4 (whether a trusted display mitigates authorisation substitution) is not yet answerable: the display is specified but not implemented. RQ5 (whether a defensible evaluation is possible using only on-chip instrumentation) remains a methodological argument, made in Section 3.7, awaiting the empirical test that would settle it.

Hypothesis H1 combines a TVLA condition and a latency condition. The latency condition, below 25 ms, is currently false as built: the measured 45.0 ms exceeds it. The TVLA condition is untested. H1 is therefore not confirmed by the present evidence, and is reported as such rather than deferred silently. Its latency clause is expected, on the basis of Table 4.5, to become achievable once the radix-4 or radix-8 multiplier recommended in Section 5.7 is implemented; its TVLA clause awaits the on-chip sensor.

5.5 Contributions of the Study

- **A hardware-verified, area-efficient signing core.** Constant-time complete-formula point arithmetic and a Montgomery ladder, synthesised at 40.4% of a 20,800 LUT device with timing closed at 50 MHz, corroborating the feasibility argued from the literature in Section 2.6 with a direct measurement on the same class of device rather than a projection.
- **A reference-model methodology that cannot silently diverge from the hardware it verifies.** The point-arithmetic instruction list is authored once, in the model, and emitted directly as the hardware ROM, so the two share one definition. The same methodology caught a real timing defect in scalar blinding before any hardware existed, reported in full in Section 4.8 rather than quietly corrected.
- **An honest account of a platform deviation, argued rather than hidden.** Section 3.2 reports the change from the specified Ultra96-V2 to a smaller, cheaper device as a factor shaping every subsequent design decision, and Chapter 4 shows the resulting design still fits with room to spare.
- **A precisely attributed and costed performance gap.** Rather than a missed target discovered by a reader, Section 4.7 traces the latency shortfall to a specific architectural cause, the bit-serial multiplier, and costs the fix using resource evidence, ninety unused DSP48E1 slices, already measured on the same device.
- **A self-checking hardware demonstration with a built-in negative control.** demo_top (Section 3.16) does not merely display a result; the fault-injection button demonstrates, on demand, that the pass indicator is capable of failing, which is the evidence that it is checking something rather than displaying a fixed value.

5.6 Limitations

- The results in Chapter 4 describe the signing core, `demo_top`: field arithmetic, point arithmetic, and the on-board self-test. They are not measurements of the complete ladder, the protocol controller, or a full signature, several of which are derived quantities rather than direct measurements, and this distinction is maintained throughout but is worth restating here.
- Timing closure at 50 MHz carries a slack of +0.405 ns, about two per cent of the clock period. This is closure, not comfortable margin, and any future change to the datapath, including the radix optimisation recommended in Section 5.7, should be expected to need re-closure and plausibly pipelining.
- No side-channel measurement of any kind has been taken from real hardware. Constant-time behaviour is established by cycle count and by the structural argument that no branch in the design depends on secret data; neither is a substitute for a measured power or electromagnetic trace.
- Scalar blinding and Coron re-randomisation are validated only in the reference model. Their hardware realisation depends on the outstanding protocol controller and has not itself been synthesised or tested.

- The functional test suites, 20 multiplier vectors and 24 point-arithmetic vectors, are thorough for their size and were deliberately constructed to include every degenerate case, but they are not exhaustive over a 256-bit input space, and no formal proof of correctness accompanies them.

5.7 Recommendations for Future Work

The following are recommended, in the order in which the dependencies among them suggest they be undertaken.

- Implement modular inversion in hardware, needed for the nonce inverse and for projective-to-affine conversion, the first piece the outstanding protocol controller depends on.
- Implement RFC 6979 deterministic nonce derivation and the HMAC-SHA-256 unit it requires.
- Implement the ECDSA protocol controller specified in Section 3.17, sequencing the above into a complete signature and wiring in scalar blinding and Coron re-randomisation as synthesised RTL rather than reference-model-only countermeasures.
- Implement the radix-4 or radix-8 multiplier projected in Section 4.7. This is recommended early because it directly closes the Objective 4 gap and because the resource evidence for it, ninety free DSP48E1 slices, is already in hand.
- Instantiate and calibrate the on-chip voltage-drop leakage sensor described in Section 3.7, and characterise its sensitivity against a deliberately unprotected control variant before trusting any result obtained from the protected design.
- Acquire and analyse trace populations for Correlation Power Analysis and Test Vector Leakage Assessment against both variants, with trace counts reported, to close RQ1 and the TVLA clause of H1.
- Implement address randomisation within the conditional swap, mitigating address-bit differential power analysis as identified in Section 3.4.1.
- Implement the HDMI display controller and physical confirmation path specified in Section 3.19, and demonstrate the What-You-See-Is-What-You-Sign property under a simulated host compromise, to close RQ4.
- Carry out an end-to-end demonstration: construct a test-network transaction, sign it on the device, and broadcast the result to a public test network.

5.8 Concluding Remarks

The problem this study set out to address, that key confidentiality and authorisation integrity are separate properties and commercial hardware wallets have repeatedly failed on one or the other, remains as real at the end of this work as at its beginning. What has changed is that a specific, costed, hardware-verified answer to the harder of the two halves of that problem, a constant-time signing core small enough for a device costing a fraction of the platform originally

specified, now exists and has been measured rather than assumed. It does not yet sign a transaction, and it does not yet render one. What it does is establish, with evidence rather than a projection, that the core arithmetic the rest of the system depends on is both fast enough to be worth finishing and small enough to leave room for the parts that would finish it.

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